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THE UNEASY CASE FOR COPYRIGHT: A STUDY OF COPYRIGHT IN BOOKS, PHOTOCOPIES, AND COMPUTER PROGRAMS

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Congress is currently considering the first major revision of the Copyright Act of 1909. Professor Breyer examines the moral and economic rationale for copyright in books. He goes on to consider proposals that would lengthen the term of protection and increase its scope in relation to photocopies and computer programs. On the basis of existing evidence he is unable to conclude that copyright should be abolished, but he argues that its extension is unnecessary and would be harmful.

I. INTRODUCTION

MACAULAY'S statement that copyright was "a tax on readers for the purpose of giving a bounty to writers"¹ reveals the conflict of interest between the reader and the book producer that underlies much of the discussion about copyright law.² The reader wants cheap books. He knows that if the copyright law did not provide an exclusive right to manufacture and sell a particular title,³ and thereby prevent competition in publishing each book, some book prices would fall, making the book more readily available. Moreover, he notes that once the ideas or language that a book contains have been created, they are no more scarce or costly than sunshine or air; and he believes that the public should be able to obtain them easily.⁴ Although the reader accepts the need for payment to induce the author and publisher

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¹ T. MACAULAY, *SPEECHES ON COPYRIGHT* 25 (C. Gaston ed. 1914) [hereinafter cited as MACAULAY].

² Compare, e.g., Plant, *The Economic Aspects of Copyright in Books*, 1 *ECONOMICA* 167 (new series 1934) [hereinafter cited as Plant], and Hurt & Schuchman, *The Economic Rationale of Copyright*, 56 *AM. ECON. REV.*, May, 1966, at 421 (1965 Papers and Proceedings of the Amer. Econ. Ass'n), with M. NIMMER, *NIMMER ON COPYRIGHT* §§ 3.1, 5.4, 48 (1970) [hereinafter cited as NIMMER]. Plant and Hurt & Schuchman contain useful discussions of the economics of copyright.

³ The law, in fact provides an exclusive right "to copy." 17 U.S.C. § 1(a) (1964). This right, in effect, allows the author (and the publisher who takes an exclusive license from the author) to exclude others from the business of making and selling books.

⁴ See the separate report of Sir Louis Mallet, C.B., in *ROYAL COMM'N ON COPYRIGHT, THE ROYAL COMMISSIONS AND THE REPORT OF THE COMMISSIONERS* (London 1878) p. xlvii [hereinafter cited as Mallet]. Mallet added:

to produce a book, he fears that the copyright allows its owner to charge considerably more than is necessary.

Book producers, on the other hand, claim that copyright protection is crucial to insure adequate book production. Without copyright protection a copying publisher could avoid many of the costs of the original publisher — payment to the author, editing, plate composition, even initial advertising — simply by photographing the printed pages of a published book. If competition then forced book prices down to the copier's cost, the first publisher and the author could not obtain adequate compensation.

A copyright law, as Macaulay also points out, may represent one way of resolving the conflict between the need for book revenues high enough to secure adequate production and book prices low enough not to interfere with widespread dissemination of what is written.⁵ Such a law protects a writer's "expression" but allows free use of the "ideas" that his work contains.⁶ A copyright system, however, is not the only way to resolve this conflict. It would be possible, for instance, to do without copyright, relying upon authors, publishers, and buyers to work out arrangements among themselves that would provide books' creators with enough money to produce them. Authors in ancient times, as well as monks and scholars in the middle ages, wrote and were paid for their writings without copyright protection.⁷ In the nineteenth century American publishers sold countless copies of British works and paid their authors royalties despite the fact that

Copies of such works may be multiplied indefinitely, subject to the cost of paper and of printing which alone, but for copyright, would limit the supply, and any demand, however great, would be attended not only by no conceivable injury to society, but on the contrary, in the case of useful works, by the greatest possible advantage.

Id. at xlviii.

As Mallet's statement suggests, writing resembles what many economists refer to as "public goods." According to Professor Paul Samuelson, a public good may be defined as something whose consumption by one man "does not diminish the quantity available for another." Samuelson, *The Pure Theory of Public Expenditure*, 36 *REV. OF ECON. & STAT.* 387-89 (1954); accord Davis & Whinston, *On the Distinction Between Public and Private Goods*, 57 *AM. ECON. REV.*, May, 1967, at 360 (1966 *Papers and Proceedings of the Amer. Econ. Ass'n*). National defense is a prime example of a public good.

⁵ See MACAULAY 21-23.

⁶ See NIMMER § 143.11; B. KAPLAN, *AN UNHURRIED VIEW OF COPYRIGHT* 32, 47 *et seq.* (1967) [hereinafter cited as KAPLAN]. In America, with numerous exceptions of varying degrees of intricacy, see, e.g., NIMMER § 145, the law requires the owner's permission before protected works can be copied and before some works such as musical compositions can be used. A radio station, for example, must obtain permission to play (for profit) a record containing a copyrighted song; a library need not obtain permission to lend (even for profit) a copyrighted book. See 17 U.S.C. § 1 (1964).

⁷ G. PUTNAM, *AUTHORS AND THEIR PUBLIC IN ANCIENT TIMES* 140, 200, 224-25 (2d rev. ed. 1894); 2 G. PUTNAM, *BOOKS AND THEIR MAKERS DURING THE MIDDLE AGES* 478-79 (1896).

American copyright law did not protect British works.⁸ It would also be possible for the government to pay for books to be produced.⁹

Nonetheless, reliance upon copyright protection is widespread. In fact, throughout recent lengthy congressional hearings on a major Copyright Revision Bill,¹⁰ the wisdom of such protection was often taken for granted. But copyright protection has faults. And its drawbacks should be more thoroughly understood before Congress votes on a bill that would significantly extend its boundaries.

Part II of this article will explore the assumptions and the principles underlying copyright protection; it will ask what would happen to book publishing if copyright were abolished.¹¹ We shall conclude that the case for copyright in the book trade is not

⁸ Mallet at 1; Response of Herbert Spencer to Question 5269, ROYAL COMM'N ON COPYRIGHT, MINUTES OF THE EVIDENCE TAKEN BEFORE THE ROYAL COMMISSION ON COPYRIGHT 287 (1878) [hereinafter cited as BRITISH MINUTES 1878]. Until 1891 this country provided no legal protection for the published works of foreigners. KAPLAN 123.

⁹ At present the federal government pays for nearly two-thirds of all research and development activity in the United States. ORGANIZATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT (OECD), REVIEWS OF NATIONAL SCIENCE POLICY, UNITED STATES 34 (1968). See notes 28, 100, *infra*.

¹⁰ The Copyright Revision Bill, S. 543, 91st Cong., 1st Sess. (1969), represents the culmination of at least fifteen years work by the Register of Copyrights and other interested parties. The Copyright Office produced a series of extensive studies (now collected in THE COPYRIGHT Soc'y OF THE U.S.A., STUDIES ON COPYRIGHT (Arthur Fisher mem. ed. 1963)). Representatives of the office then met with representatives of affected interest groups. The bill was originally introduced in the House where hearings were held in 1965. See *Hearings on H.R. 4347, H.R. 5680, H.R. 6831, H.R. 6835 Before Subcomm. No. 3 of the House Comm. on the Judiciary*, 89th Cong., 1st Sess. (1965). The House passed the bill, but the Senate did not report it out of committee. The bill was reintroduced in this session of Congress and has been reported out of a Senate subcommittee. It is currently before the Senate Committee on the Judiciary. For a more complete history of the bill, see Nimmer, *Two Copyright Crises, Foreword to Project — New Technology and The Law of Copyright, Reprography and Computers*, 15 U.C.L.A. L. REV. 931 (1968).

¹¹ This discussion will focus primarily upon books despite the fact that books now constitute a small portion of the products that copyright protects in this country. Those products include photographs, dress designs, films, musical compositions, architects' drawings, and various other creations. See NIMMER §§ 10-38. But originally copyright was created to answer book publishers' problems. See generally KAPLAN 2-9; M.-C. DOCK, *ETUDE SUR LE DROIT D'AUTEUR* 63-75 (1963) [hereinafter cited as DOCK]. And book publishing as now organized well illustrates the case for copyright.

Moreover the article will discuss statutory copyright protection which is granted from the time a work is published. In America a "common law" of copyright prohibits the unauthorized copying of writings from the time of their creation until they are published. I shall assume that in order to protect the privacy of authors, the law ought to prohibit most unauthorized copying of writings prior to their initial publication.

a strong one generally and is even weaker for some parts of that trade.¹² Taken as a whole, the evidence now available suggests that, although we should hesitate to abolish copyright protection, we should equally hesitate to extend or strengthen it.

Part III of the article will examine several ways in which the comprehensive Copyright Revision Bill now before Congress would increase protection.¹³ The bill would extend copyright's duration from a maximum of fifty-six years to "life of the author plus fifty years."¹⁴ It would allow more effective protection against those who make xeroxed¹⁵ copies of books. It would allow the copyrighting of computer programs. Part III will argue that in none of these areas has the case for increased protection been adequately made.

II. THE BASIC ARGUMENTS FOR PROTECTION

Arguments for copyright protection are of two sorts: Some are based upon the author's "moral rights" to reap the fruit of his labors or to control what he has created. Others rest upon the economic inducement to publication that copyright provides. We must analyze both lines of argument before reaching conclusions about the value of existing copyright law.

A. "Moral" Justifications for Copyright Protection

The theory that authors have a natural right to the fruit of their labors is an ancient one. In the sixteenth century a noted French lawyer argued that a book cannot be printed without the author's permission, for just as "the heavens and the earth belong to [God], because they are the work of his word . . . [s]o the author of a book is its complete master, and as such can dispose of it as he chooses."¹⁶ A similar theory probably under-

¹² This conclusion rests in part upon economic information that is far from complete. To quote the *Economist*, "for an industry whose product is the printed word, the book trade has strangely little information on record about its own fortunes." Quoted in B. FINDLATER, *WHAT ARE WRITERS WORTH?* 1 (1963). Despite volumes of studies and testimony in preparation for the Copyright Revision Bill, we still have little concrete information about how authors respond to changes in prices. Yet in analyzing legal policy, I believe it preferable to do one's best with the facts available than to write a more limited analysis that ignores them or to refrain from drawing policy conclusions on the ground that better information is needed.

¹³ See note 10 *supra*.

¹⁴ Currently the original copyright protection runs for 28 years, with a renewal period of 28 more years open to the author or certain relatives or associates. 17 U.S.C. § 24 (1964).

¹⁵ Though arguably "reprography" is the generic term to refer to reproduction methods such as "xeroxing," I shall use the latter term, with which the reader is more familiar.

¹⁶ The lawyer was Marion and his argument is reprinted in Dock 78.

lies the legendary King Diarmed's succinct, bucolic analysis of copyright: "To every cow its calf."¹⁷ The Register of Copyrights enlarges on this view when he claims that the law "has an important secondary purpose: To give authors the reward due them for their contribution to society."¹⁸ The difficulty with the theory lies in discerning the extent to which an author should be able to maintain control of a published book. How large is the fruit? How big the calf? Just what is the reward that is due him?

It is unlikely that "fruit" or "reward" is meant to refer to the amount of money needed to persuade a man to write a book. One has no need for a copyright law in order to guarantee the payment of such a sum, for, in the absence of slavery, an author will not write unless he is paid his "persuasion" cost. "Fruit" or "reward," then, must mean something more.

It is often said that an author should receive the "value" of his work to society — a value that might be measured in terms of what those who benefit from the book would be willing to pay rather than do without it.¹⁹ But few workers receive salaries that approach the total value of what they produce. The social value of the work performed by the man who invents the supermarket, the man who clears a swamp, the academic scientist, or the school-teacher may be much greater than his pay.²⁰ Moreover, workers in competitive industries make products that sell at prices well below what many of their buyers would be willing to pay for them.²¹ We do not feel that owners, managers, or workers in such

¹⁷ A. BIRRELL, SEVEN LECTURES ON THE LAW AND HISTORY OF COPYRIGHT IN BOOKS 42 (1899).

¹⁸ *Report of the Register of Copyrights on the General Revision of the U.S. Copyright Law for House Comm. on the Judiciary*, 87th Cong., 1st Sess. 5 (Comm. Print 1961).

¹⁹ Of course, to measure a book's total social value is exceedingly difficult. The ideas that a book contains (a theory of relativity, for example) may prove of enormous, and differing, value to persons who do not buy a book. The book's publication will also benefit in differing amounts those who sell library equipment or other book supplies to readers. The book itself will differ in value to different readers. Existing copyright law allows the owner to charge only a book's readers for the value they receive from it and those readers often pay a uniform price. For the law to attempt to collect varying amounts of money from the different beneficiaries would impose an enormous administrative cost. But it may nonetheless prove administratively easy to give the author some, if not all, of the social value of what he creates (over and above his persuasion costs).

²⁰ The wages paid these workers do not reflect the extent to which they benefit society, for many of those who benefit from their work pay them nothing. Where such "spillover" benefits exist, we may wish to raise wages in order to attract more people into these particular occupations. Such a reason for raising wages, however, is economic, not moral.

²¹ In a competitive market with no price discrimination possible, the total return to all factors equals the price times the number of units sold. The value of the product, however, is (roughly speaking) the sum of the price each user would

industries are for this reason morally entitled to higher wages.²² Indeed, when a worker without competition — perhaps because he is the only doctor in the area, or the only engineer capable of building a certain bridge — could charge a price close to the total value of his services to the buyer, we normally encourage competition, which will force him to charge less.

There is nothing inherently immoral in the fact that many workers are paid less than the social value of what they produce, for much of the excess of social value over persuasion cost is transferred to the consumer in the form of lower prices.²³ It is not apparent that the producer has any stronger claim to the surplus than the consumer or that the author's claim is any stronger than that of other workers. In fact, why is the author's moral claim to be paid more than his persuasion cost any stronger than the claim of others also responsible for producing his book: the publisher, the printer, the bookseller, and those responsible for the literature of the past that inspired him?

In sum, simply to speak of "fruits of one's labor" does not show that the author should be paid more than his persuasion cost or how much more he should be paid. In particular, it does not demonstrate that the amount he receives under existing copyright law is any more "just" than what he would receive without copyright protection or under a different copyright system.²⁴

Two special claims might be advanced, however, on the author's behalf, though neither persuades me of the need for copyright. First, one could argue that society, regardless of how it treats other workers, should especially reward a creator of a great work of lasting social value for the general contribution he has made to social progress, a contribution that extends over and beyond the value he has provided purchasers of his work. But this claim, even if accepted, provides only limited support for copyright. For one thing, a copyright system on this theory rewards the wrong works. It tends to favor the commercial, popular work with large sales, for the buyers of many serious and important

pay rather than do without the product. Since some users would pay much more than the going price rather than go without the product, the total value of the product can be considerably greater than the total return to all of its factors.

²² In such a case, workers are not creating "spillover" benefits; there is no obvious economic (or moral) argument for raising their wages.

²³ The difference between the price a consumer is willing to pay to obtain a good and the price he actually pays is often called "consumer's surplus." A monopolist who is able to charge a different price for each unit of output (discriminating perfectly in price) can extract all the consumer's surplus for his own benefit. Competition and other impediments to perfect price discrimination will prevent him from doing so, and will lead to lower prices for many (and sometimes for almost all) consumers. See J. HICKS, *VALUE AND CAPITAL* 38-41 (2d ed. 1946).

²⁴ As the remainder of Part II will suggest, to abolish copyright protection would not necessarily drive returns to the author down to his "persuasion cost."

works are often few in number²⁵ and notoriously in need of funds.²⁶ Further, copyright's tendency, through higher prices, to restrict dissemination will prove particularly harmful when the work is a great one. Finally, a more equitable distribution of the cost burden, and one that will widen the dissemination of serious works, can be devised without great difficulty in the form of subsidies, grants, or prizes from the government, foundations, or universities. It is not unfair to finance through taxes the creation of works that benefit not only those who buy them but also many other members of society as well. And such a system can channel as much or more money to the author as would a copyright system, but without simultaneously restricting the dissemination of his work.²⁷ The practicality of administering such a system — when restricted to works of particular social value — is suggested by the fact that many governments and other institutions now award such grants and prizes. In the United States the amount of such support is large when compared with the total revenue that scholarly, technical, or scientific writers receive from copyright royalties, and even in the case of literary works it is significant.²⁸

²⁵ Herbert Spencer's *SOCIAL STATICS* (1851), for example, was first published in an edition of 750 copies; that edition took fourteen years to sell. H. SPENCER, *VARIOUS FRAGMENTS* 20 (1898).

²⁶ That the financial plight of orchestras and other performing groups is very serious and growing worse is well documented in W. BAUMOL & W. BOWEN, *PERFORMING ARTS — THE ECONOMIC DILEMMA* (1966).

²⁷ Suppose, for example, that Jones' book of poems is sufficiently good to entitle him "morally" to \$5000. Suppose further that he can earn that \$5000 by selling his book for \$10 per copy. Were Jones to receive the \$5000 instead from the government or a foundation and were copyright restrictions removed from his book, its price might fall while leaving him no worse off financially. Readers who could not afford to pay \$10 for the book are better off, and Jones obtains a wider circulation. Whether this sort of virtue can be retained in any widespread, practically administrable, subsidy scheme, however, is doubtful. See note 100 *infra*.

²⁸ If we multiply total book publishing receipts of \$2,125,100,000 (U.S. DEPT OF COMMERCE, BUREAU OF THE CENSUS, 1967 CENSUS OF MANUFACTURES, INDUSTRY SERIES, BOOK PUBLISHING 3 (Preliminary Report Oct. 1969)) [hereinafter cited as 1967 CENSUS OF MANUFACTURES], by the 10% to 15% frequently mentioned as the average royalty payment (see, e.g., *Royalties: Current Patterns and Practices*, PUBLISHERS' WEEKLY 54 (Feb. 15, 1965) and *Royalties — III: Current Practices Affecting Scientific and Technical Books*, PUBLISHERS' WEEKLY 65 (April 19, 1965)) authors' royalties amount to between \$200 and \$300 million. In the case of adult tradebooks (both paperback and hardbound) total publishers' receipts (not counting book club sales) amount to about \$216.7 million (1967 CENSUS OF MANUFACTURES 3). Authors' royalties from this source amount to approximately \$35 million. ERNST & ERNST, 1968 FINANCIAL & OPERATING RATIO REPORT TO AMERICAN BOOK PUBLISHERS COUNCIL 9-9a [hereinafter cited as ERNST & ERNST]. (I would like to thank the Council for allowing me to use the survey for this article.) These authors may earn up to \$24 million from the sale of subsidiary rights to book clubs, movie makers, etc. See W. LORD, *HOW AUTHORS MAKE A LIVING* 112 (1962)

The second claim for special consideration for authors rests upon an intuitive, unanalyzed feeling that an author's book is his "property." But why do we have such a feeling? An intellectual creation differs radically from land and chattels. Since

[hereinafter cited as LORD]. This would increase total royalty income to roughly \$60 million. See notes 116, 117 *infra*.

Estimating the amount of government and other institutional support of authors for contrast is not easy because of the various forms that support may take. For example, the federal government in 1965 spent \$13 billion on research and development including \$1.8 billion for fundamental research. ORGANISATION FOR ECONOMIC CO-OPERATION AND DEVELOPMENT, REVIEWS OF NATIONAL SCIENCE POLICY: UNITED STATES 34 (1968). Presumably a considerable amount of this money was paid for published work of many scientists. Moreover, universities support much writing by paying the salaries of their teachers who write.

Some notion of the extent of foundation support for literature and the humanities may be obtained by considering the fact that, in 1962, 6,000 foundations provided about \$40 million for the humanities and 53 foundations dispersed \$3 million under the general heading of "Language and Literature." Dempsey, *The Foundations: A Welfare State for Writers?* HARPERS 165-69 (Oct. 1965). The Ford Foundation gives about \$9 million annually for its Humanities and Arts Program and in 1962 about \$2 million in Guggenheim fellowships were awarded to scholars. *Id.* at 169.

Grants and prizes specifically earmarked for creative writing, however, may be considerably smaller. Of the 354 Guggenheims awarded to scholars in 1962, for example, 11 went to creative writers. *Id.* at 166. LITERARY MARKET PLACE 557-59 (1969-70) lists approximately an additional \$100,000 in specified grants. The National Endowment for the Arts reports granting \$737,010 for literature in 1967, which includes \$205,000 in direct grants to creative writers. NATIONAL ENDOWMENT FOR THE ARTS AND NATIONAL COUNCIL ON THE ARTS, ANNUAL REPORT FOR THE FISCAL YEAR ENDED JUNE 30, 1967, at 56. The National Endowment for the Humanities reports spending \$300,000 to support the publication of definitive editions of American literary "classics." NATIONAL ENDOWMENT FOR THE HUMANITIES, THIRD ANNUAL REPORT, FISCAL YEAR 1968, at 15. American authors may receive nearly \$200,000 per year in general literary prizes (which includes several international prizes for which they are eligible); publishers make available approximately an additional \$200,000 in prize money. LITERARY MARKET PLACE 533-57 (1969-70). Of course, many creative writers are subsidized by universities which employ them to teach. See *Trusts and True Foundations*, 1965 TIMES LITERARY SUPPLEMENT 1057-58.

Foreign countries, too, offer forms of subsidy. France, for example, awards 475 literary prizes. *A Multitude of Awards*, 1961 TIMES LITERARY SUPPLEMENT 733. The American Society of Composers, Authors and Publishers also awards special prizes to "writers whose works have a unique prestige value for which adequate compensation would not otherwise be received." The awards' recipients are selected by panels of distinguished experts. Finkelstein, *Public Performance Rights in Music and Performance Right Societies*, in 7 COPYRIGHT PROBLEMS ANALYZED 19 (rev. ed. 1961).

Of course, a system of copyright protection *together with* grants and prizes may channel more money to the creators of great works than either system alone. Yet, in view of the fact that great works constitute a small proportion of all works protected by copyright (and in view of the considerations mentioned above), I cannot accept this reason for granting copyright protection as a critically important one.

ideas are infinitely divisible, property rights are not needed to prevent congestion, interference, or strife. Nor does the fact that the book is the author's *creation* seem a sufficient reason for making it his *property*.²⁹ We do not ordinarily create or modify property rights, nor even award compensation, solely on the basis of labor expended.³⁰

What may lie at the root of this intuition is a notion that ~~property rights~~ are often created for reasons of efficiency. To provide a man with property rights in land and to give him the right to enjoy profits from the sale of crops may encourage him to care for the land, putting it to its most productive use. Providing him with this right may thereby allow the earth's resources to satisfy more human wants.³¹ Such a justification supports copyright protection, however, only if copyright is a more efficient way than other feasible institutional arrangements to satisfy the human want for writings. This is not a moral judgment, but an economic one. And if copyright is more efficient — a question dealt with in section B *infra* — that fact alone provides a strong reason for adopting it without relying upon analogies to other property rights.³²

Another noneconomic claim for copyright asserts that the law is needed to protect important personal interests of the au-

²⁹ One might argue along Lockean lines that "whenever one mingles his effort with the raw stuff of the world, any resulting product ought — simply ought — to be his." Michelman, *Property, Utility, and Fairness: Comments on the Ethical Foundations of "Just Compensation" Law*, 80 HARV. L. REV. 1165, 1204 (1967). J. LOCKE, *THE SECOND TREATISE OF CIVIL GOVERNMENT* §§ 27-30 (P. Lasslett ed. 1960). But the difficulty with this theory is that it neither explains *why* property ought to be created on this basis nor does it describe our actual practice. See note 30 *infra*.

³⁰ Countless governmental actions (as well as private actions) continuously change property values without reference to human labor. By enriching some persons and impoverishing others, they prevent wealth or income from remaining proportional to the amount of work put forth. For example, we allow collective bargaining; we set minimum wages; we prohibit traffic in dangerous drugs; we forbid billboards along highways. Such actions affect the value of both labor and property.

³¹ See J. BENTHAM, *THEORY OF LEGISLATION* 115-19 (New & Cheaper ed. 1905); Demstet, *Toward a Theory of Property Rights*, 57 AM. ECON. REV., May, 1967, at 347 (1966 Papers and Proceedings of the Amer. Econ. Ass'n).

³² Does the human desire for property, including intellectual property, also reflect a "territorial" instinct? Matthew Arnold pointed out that man has "a strong instinct making him seek to possess what he has produced or acquired." Yet Arnold also noted that men have tended to embody that instinct in law only "so far as may be consistent with the general advantage of the community." 11 M. ARNOLD, *THE WORKS OF MATTHEW ARNOLD, IRISH ESSAYS AND OTHERS* 241-42 (1904). References to instinct will not determine whether a copyright law works to that "general advantage."

thor.³³ In the eighteenth century Immanuel Kant urged that an unauthorized copier might garble an author's work, misrepresent it, or put into circulation a work that the writer did not wish to present to *this* public at *this* time. The author might then be put to the considerable trouble of explaining to the public what he in fact wrote or meant by what he wrote.³⁴ Kant probably saw the author's attempt to communicate with the public as an *act* of the author with which unauthorized copying could unjustifiably interfere; because such copying might thus unnecessarily inhibit the author's liberty, he saw a "natural obligation" not to "counterfeit" books.³⁵

One might quarrel with Kant's argument, for the fact that unauthorized copying may *sometimes* interfere with an author's personal interests does not show that it *always* (or even *often*) will do so. One suspects that many authors dislike copying because of the injury to their purse, not their pride. But in any event, to accept the need for protecting such personal interests of the author does not require accepting a system of copyright law. In fact, American copyright law does not protect these interests directly. Rather, by forbidding unauthorized reproduction, it provides a protective umbrella beneath which author and publisher can work out, *in their contract*, safeguards for the author. One might instead give the author's interests explicit statutory recognition, as is done in many European countries,³⁶ or one might allow common law courts to protect them by applying principles of tort law. Such principles would allow courts to weigh the dignitary interests of the author against the publisher's or producer's interest in, for example, publishing anonymously, or altering the text, or inserting TV commercials.³⁷

Tort protection for the author is likely to be at least as gen-

³³ These interests include: 1) The author's right to have his work acknowledged; 2) his right not to have his work attributed to someone else; 3) his right to have his work presented free of garbling; 4) his right (recognized in certain European countries) to withdraw his work from circulation when, for example, he changes his mind about the truth of what it says. See Treece, *American Law Analogues of the Author's "Moral Right,"* 16 AM. J. COMP. L. 487 (1968).

³⁴ The unauthorized publication in the West of the works of certain Russian authors may provide cases in point. See, e.g., Alexander Solzhenitsyn's letter to LITERATURNAYA GAZETA (April 21, 1968) asking that his book CANCER WARD not be published in the West without his permission. Reprinted in A. SOLZHENITSYN, CANCER WARD 558-59 (Bantam ed. 1969).

³⁵ See I. KANT, *Of the Injustice of Counterfeiting Books*, in 1 ESSAYS AND TREATISES ON MORAL, POLITICAL, AND VARIOUS PHILOSOPHICAL SUBJECTS 225, 229-30 (W. Richardson trans. 1798).

³⁶ See, e.g., Sarraute, *Current Theory on the Moral Right of Authors and Artists Under French Law*, 16 AM. J. COMP. L. 465 (1968). The rights protected are normally those listed in note 33 *supra*.

³⁷ Cf. *Stevens v. National Broadcasting Co.*, 148 U.S.P.Q. 755 (Cal. Super. Ct. 1966).

erous as that now provided. The "interest balancing" approach of courts in tort cases strongly resembles the way in which courts now interpret unspecified intent in a publishing contract.³⁸ While an author might contract away tort protection, so can he now choose to forego the protection that a contract could provide. In fact, tort protection should prove somewhat more generous, for unlike copyright it would apply to works in the public domain.³⁹

Moreover, tort law in this area, while undeveloped, nonetheless exists. There is tort precedent that prohibits garbling a man's work and then attributing it to him⁴⁰ and precedent that provides some protection for a work's integrity.⁴¹ Despite complexities, there seem no serious obstacles to developing a comprehensive system of tort protection. Probably, without copyright, protection for the author's dignitary rights would still exist, albeit in a new form.

* * *

In sum, none of the noneconomic goals served by copyright law seems an adequate justification for a copyright system. The system provides a fairly useful way to protect an author's dignitary rights, and it can provide extra compensation to some writers. Yet alternative legal arrangements can readily be devised to protect the personal nonpecuniary interests of the author. Moreover, the merits of the claim to extra compensation have not been clearly established, nor has copyright proved particularly effective in providing compensation for the "great work" of unusual social value. If we are to justify copyright protection, we must turn to its economic objectives.

B. The Economic Arguments

To explore the nature and the strength of the economic case for copyrighting books we must ask what would happen if copyright protection were abolished. Would abolition seriously threaten book production? Which books would it threaten most, and why? What is to be gained from abolition? Who would benefit, and how? Our social cost-benefit analysis will not be

³⁸ The courts weigh the strength of the author's interest against the reasonableness of the publisher's behavior. *See id.*; Treece, *supra* note 33, at 494-500.

³⁹ It is also possible that tort protection will lead courts to treat authors somewhat more generously than they are now treated when contracts are silent. *See Vargas v. Esquire, Inc.*, 164 F.2d 522 (7th Cir. 1947) (artist denied relief when nude portraits he drew under contract were published without his name). Such protection may make authors somewhat less dependent upon the vagaries of specific contractual language.

⁴⁰ *See Drummond v. Altemus*, 60 F. 338 (C.C.E.D. Pa. 1894).

⁴¹ *See Preminger v. Columbia Pictures Corp.*, 148 U.S.P.Q. 398 (N.Y. Sup. Ct.), *aff'd*, 149 U.S.P.Q. 872 (N.Y. App. Div.), *aff'd per curiam*, 150 U.S.P.Q. 829 (N.Y. Ct. App. 1966); *Stevens v. National Broadcasting Co.*, 148 U.S.P.Q. 755 (Cal. Super Ct. 1966).

precise, for there are not enough facts readily available to make exact quantitative estimates. Nonetheless, one can identify the *sources* of benefit and loss, estimate whether the amounts involved are significant, and draw approximate conclusions about whether copyright protection seems sufficiently valuable to justify not only retaining it, but also extending its scope.

Much of what follows will discuss copyright in terms of the protection it provides not to an author, but to a publisher — a protection that flows from the exclusive license to publish that an author normally grants him. Historically the publisher led the fight for laws that allow him to obtain exclusive rights. After the invention of printing the first state grants of exclusive rights — the precursors of modern copyright protection — were made (in Venice in 1495 and in France in 1507⁴²) to printers,⁴³ not to authors. In the sixteenth⁴⁴ and seventeenth⁴⁵ centuries as well as today⁴⁶ the publisher has argued that not only the author's, but also his own, considerable investment in bringing a book to market must be protected. To emphasize the publisher does not slight the author, for as long as a publisher believes he can make money from a book its author can bargain for a share of the publisher's revenue by threatening to withhold it or go elsewhere.⁴⁷ This is not to say that publishers' and authors' interests always coincide, but where divergences are of importance to the argument, they will be pointed out.

The discussion will also distinguish, as publishers do, among the different submarkets within publishing. *Textbooks* account

⁴² There is debate whether the first Venetian grant was made in 1469 to Jean de Spire or in 1495 to the printer Alde Manuce. The first French grant was made in 1507 to Antoine Verard for an edition of the Epistles of St. Paul annotated by a member of the Theology Faculty. Dock 63-64. The first English grants of an exclusive right to print came in 1518. Later, partly as an aid to royal censorship, a guild of publishers, the Stationers' Company, was given the exclusive privilege of printing books in England for almost 150 years. See L. PATTERSON, *COPYRIGHT IN HISTORICAL PERSPECTIVE* 20-120 (1968).

⁴³ Today's publisher is normally not a printer. He acts as an editor and broker, obtaining manuscripts, arranging for their printing, and securing their distribution. The early printers, however, themselves engaged in many of these activities. See L. PATTERSON, *supra* note 42, at 45-46.

⁴⁴ See Dock 62-66; KAPLAN 3; L. PATTERSON, *supra* note 42, at 20-150 *passim*, 222-29.

⁴⁵ See, e.g., KAPLAN 5-6; Plant 176-78 (petition of Stationers' Company to Parliament in 1642).

⁴⁶ See, e.g., Markham, *An Economic Analysis of the New Information Technologies*, in AMERICAN TEXTBOOK PUBLISHERS' INSTITUTE & AMERICAN BOOK PUBLISHERS COUNCIL, *AN ECONOMIC-MEDIA STUDY OF BOOK PUBLISHING* 1-27 (1966) [hereinafter cited as Markham].

⁴⁷ As long as the author retains his common law copyright (as I believe he should), see note 11 *supra*, a publisher will be unable to publish a book (not previously published) without the author's permission. See NIMMER § 11.

for the single largest portion (about thirty-five percent) of all publishing revenues.⁴⁸ They comprise books used in colleges and books used in elementary or high schools (known in the industry as "elhi" texts).⁴⁹ *Tradebooks* include the novels and popular nonfiction that ordinarily spring to mind when copyright is discussed. Adult tradebooks (both hardbound and paperback) sold by booksellers account for only about ten percent of all publishing revenue. If we add in children's books, tradebooks sold through book clubs, and pulp novels,⁵⁰ the figure rises to about twenty-seven percent.

Most works of technical or serious nonfiction (as opposed to popular nonfiction) would not ordinarily be considered tradebooks.⁵¹ For purposes of reporting sales figures to trade associations, the publisher of a scholarly book of economic theory, for example, would probably consider it a "college text" if it is used for college instruction and sold primarily to a college audience. Only if it has a large enough appeal to a general reading public to warrant the extra cost of distribution through ordinary trade channels⁵² might it also be considered a tradebook.⁵³

I shall deal principally with the two largest portions of the book market, textbooks and tradebooks, though much of what I say about textbooks will apply to scholarly nonfiction as well.

1. *Would the Abolition of Copyright Protection Seriously Injure Book Production?*

To abolish copyright protection would allow publishers to compete in the production and sale of an individual title. The proponents of copyright protection would argue strongly that

⁴⁸ 1967 CENSUS OF MANUFACTURES 3.

⁴⁹ College textbook revenue accounts for about 40% of textbook publishing revenue; elhi textbook revenue accounts for about 60%. *Id.*

⁵⁰ These are "wholesaled paperbound books usually of digest size (4 $\frac{1}{4}$ " x 6 $\frac{1}{2}$ ") and distributed 50 percent or more through magazine wholesalers." *Id.*

⁵¹ They will be considered tradebooks only if they appeal to a general audience and are sold through the same trade channels as are books of fiction and popular nonfiction. Since tradebooks are distributed to retailers at a 43% discount while textbooks are distributed at only a 20% discount, a company's textbook division will handle a scholarly book unless its predicted sales to the general public are unusually high.

⁵² The use of distribution channels as a way to distinguish between classes of books was stressed in an interview with an official of the American Educational Publishers Institute, October 1, 1970. (The Institute, a trade association of textbook publishers, has recently merged with the American Book Publishers Council, a tradebook publisher trade association. For clarity of source, however, the associations and their material will be identified by their former names.)

⁵³ The remainder of publishing revenue is accounted for by sales of hardbound technical, scientific, or professional books (11%), subscription reference books (9%), religious books (5%), and various miscellaneous items. 1967 CENSUS OF MANUFACTURES 3.

even if such competition sometimes brings the reader books at lower prices, it may often prevent him from obtaining a book at all. A subsequent publisher could copy and sell a particular book at lower costs than those incurred by the initial publisher. If competition immediately forced that book's price down to the copier's costs, the initial publisher could not recover his fixed costs or pay the author. The fear of such a result in a world without copyright would, it is claimed,⁵⁴ discourage publishers from publishing and authors from writing. To evaluate the importance of this economic claim for copyright protection, we must appraise the impact of possible copying on initial publishers' revenues and the possible harm that could flow from any such revenue loss.

(a) *Would Competition in the Production of Individual Titles Seriously Threaten Book Publishers' Revenues?*

(i) *The Copying Publisher's Cost Advantage.* — A copying publisher could produce a copy for considerably lower cost than an initial publisher, for the initial publisher incurs many costs that a subsequent publisher would not have to bear. The initial publisher must edit the book and he must pay various fixed manufacturing expenses, such as those of composing type into pages, making corrections, and designing plates for the jacket. A copying publisher can avoid many of these costs simply by photographing the pages of the completed book and using the photographs for printing purposes. He can also escape a portion of the selling, promotion, and overhead costs that an initial publisher must bear. And he might not pay the author a royalty.

Because a copier's cost advantage will vary considerably, depending upon the title and the number of copies that the book sells, one can more easily discuss the extent of the advantage in terms of specific dollar examples, rather than percentages. The following example is constructed to illustrate the typical⁵⁵ extent of the cost advantage. The example shows the differing costs to the initial publisher and to a copier of producing an unillustrated 400-page college text that is expected to sell about 5000 copies. It will be sold to booksellers for \$4.80. The booksellers will retail it for \$6.00. (The example is based upon average cost figures contained in the American Education Publishers' Institute Annual Survey of the textbook publishing industry for 1968.⁵⁶ Since textbook publishers accounting for more than ninety percent of all textbooks sold provided detailed cost information for that survey, the average cost figures are reasonably representative.)

⁵⁴ See, e.g., Markham 12.

⁵⁵ See note 56 *infra*.

⁵⁶ STANLEY B. HUNT & ASSOC., 1968 ANNUAL SURVEY OF THE TEXTBOOK PUBLISHING INDUSTRY [hereinafter cited as HUNT SURVEY]. The Institute has gra-

<i>Fixed Costs</i>	<i>First Publisher</i>	<i>Copier</i>
Plant ⁵⁷	\$4,500	\$900
Editorial ⁵⁸	1,440	500
Other overhead ⁵⁹	2,880	2,000
Selling ⁶⁰	1,000	1,000
Promotion ⁶¹	700	350
Total fixed cost	\$10,520	\$4,750
<i>Variable Costs</i>		
Manufacturing ⁶²	\$.80 per copy	\$.80 per copy
Warehouse and shipping ⁶³	.16 " "	.16 " "
Royalty to author ⁶⁴	.70 " "	.00 " "
Selling	.20 " "	.20 " "
Promotion	.14 " "	.07 " "
Publishers' total variable cost	2.00 " "	1.23 " "
Retailers' markup *	1.20 " "	1.20 " "
Total variable cost	3.20 " "	2.43 " "

* Retailers' markup is the difference between the \$6.00 price and \$4.80, which the retailer pays for the book.⁶⁵ This \$4.80 figure includes the publisher's profit.

ciously allowed me to use the figures in this study. The figures in the example and in the footnotes which accompany it are based upon the Hunt Survey and informal interviews with several publishers during which they discussed the costs of book publishing.

I have developed many of these figures by assuming that the publisher's net revenue from the sale of the book in my example would equal \$24,000 (5000 x \$4.80) and then multiplying that figure by the percentage of net revenue that the Hunt Survey states is average for a particular category. For example, the Hunt Survey states that publishers, on average, spend 6% of their net revenues on editorial work: 6% of \$24,000 is \$1440. Figures obtained in this way correspond fairly well with figures that publishers interviewed provided from their own firms' records as typical costs for producing a book. And this correspondence to some extent lessens the risks of error arising from my inability to obtain figures on the dispersion of costs around the mean.

⁵⁷ Plant costs include the cost of composing the type from which printing plates will be made, the cost of making up the plates used to print, the cost of designing and making jacket plates, the cost of corrections, and some miscellaneous expense. In this example, plant costs may break down roughly as follows:

Composition	\$3,100
Plate makeup	400
Jacket plates and design	400
Corrections	400
Miscellaneous	200
Total	\$4,500

I have assumed that a copying publisher would have to pay the cost of photographing the book, making up plates for printing the book and jacket, and that he would incur various miscellaneous expenses. Based upon my interviews with publishers, I believe the \$900 figure is a minimum. For a discussion of the technology of printing, see M. LEE, *BOOKMAKING: THE ILLUSTRATED GUIDE TO DESIGN & PRODUCTION* (1965).

The copier's cost advantage is fairly large.⁶⁰ To produce an edition of 5000 copies would cost him less than two-thirds what it would cost the initial publisher. The result is that the book's

⁵⁸ These costs include the cost of selecting manuscripts as well as revising those selected. I assume that a copying publisher will also have to read manuscripts and books to determine which he is going to copy.

⁵⁹ Overhead expenses include salaries of managers, supervisors, clerical workers, etc., as well as rent, office expenses, and similar items. I assume that a copying publisher will also incur many of these expenses. If the copier is not a well-established publisher, he may save in this area, but "fly-by-night" copiers would probably have difficulty working their books into ordinary channels of distribution.

⁶⁰ I assume that both original and copying publishers would have to work equally hard at selling their books; if anything, the copying publisher may have to spend more money to sell a book that has already been distributed to many college bookstores.

⁶¹ Promotional expenses, particularly advertising, may be more important in publishing than in many other industries, for the buyer must be told about many different books, each of which sells only a few thousand copies. Advertising expenditures for textbooks are somewhat smaller than for tradebooks, though selling expenses, which include trips to schools and other promotions, are greater.

I have assumed that half the promotional expenses are fixed and half are variable and that the copier will spend half as much on promotion as the original publisher. At present, paperback publishers, whose editions appear several months after the hardcover edition, spend about half as much on promotion; a copier would presumably do about the same.

⁶² Variable manufacturing costs include the cost per copy of paper, printing, binding, and printing the jacket. In this example, the manufacturing costs may break down as follows:

Paper	22 cents per copy
Printing	11 " " "
Binding	40 " " "
Jacket	7 " " "
Total	80 cents per copy

Both initial and copying publishers must bear this expense.

⁶³ In fact, the warehousing expense of the initial publisher may marginally exceed that of a copier because of the need to store books for a longer time.

⁶⁴ While a copier need not bear this cost, it is possible that some would pay the author a fee to obtain a copy of his manuscript before its initial publication, thereby gaining an advantage over other copiers. This practice was common in the nineteenth century. See p. 299 *infra*.

⁶⁵ In other words, the retailer will add a markup of 20% of the book's retail price. The markup for tradebooks is considerably higher, usually amounting to about 43%.

⁶⁶ A copying publisher may also run less risk than the original publisher that the book will prove unpopular. Markham assumes that the extra cost to the first publisher imposed by risk amounts to 50 cents per copy for a textbook that will sell 20,000 copies at \$6.00 each. Markham 11-12. My discussions with publishers, however, lead me to suspect that risk in the case of *textbooks* is not so large. The textbook publishers interviewed stated that they lose money on few titles. Although very profitable texts appear infrequently, they estimated that about 85% of the textbooks they published would earn some profit.

initial publisher must sell 3700 copies before he begins to earn a profit whereas a copying publisher selling the book at the same price will begin to earn a profit after selling 1400 copies. Moreover, if the copier expects to sell 2500 copies, he could set a price to the retailer of \$3.20 instead of \$4.80 and still earn a small profit. And the retailer could sell the text for \$4.40 instead of \$6.00 while earning no less than before.⁶⁷

The copier's cost advantage may be larger in the case of many complicated textbooks but smaller for many bestselling tradebooks. In general, editing, illustrations, footnotes, tables, and indices raise a book's fixed manufacturing costs, increasing the copier's advantage.⁶⁸ A large number of sales reduces that ad-

⁶⁷ I suspect that the retailer's markup in dollar terms would not fall, for even if the copier supplies the retailer with the book at a lower price, the retailer's costs of handling the book remain the same and there is no indication that retailing is not highly competitive.

⁶⁸ A copier's cost advantage may be greater in the case of dictionaries, encyclopedias, or certain elhi texts that involve considerable editorial work or other large fixed investments. The following is a more roughly worked out example of the cost of producing a 900-page elementary or high school history text, illustrated in color, which is expected to sell about 150,000 copies at a retail price of \$7.00. Because this kind of text is unusually expensive to produce, its costs are not well illustrated by average cost figures, and the example is based primarily upon discussion with publishers:

<i>Fixed Costs</i>	<i>First Publisher</i>	<i>Copier</i>
Plant	\$100,000	\$20,000
Overhead and editing	100,000	30,000
Selling and promotion (includes \$15,000 for teachers' free edition)	75,000	32,000
Total Fixed Costs	\$275,000	\$82,000
<i>Variable Costs</i>		
Manufacturing	\$1.75 per copy	\$1.75 per copy
Warehousing and shipping	.20 " "	.20 " "
Author's royalty	.75 " "	.00 " "
Promotion and selling	.50 " "	.25 " "
Publishers' total variable cost	\$3.20 per copy	\$2.20 per copy
Retailing markup	1.40 " "	1.40 " "
Total variable cost	\$4.60 per copy	\$3.60 per copy

On the other hand, the advantage may be reduced in the case of tradebooks, for the cost of retailing—an important variable cost—is much higher for tradebooks than for texts. The following example illustrates the costs of producing a 400-page hardbound tradebook (a novel, for example) expected to sell 5000 copies. The publisher will sell it to the retailer for \$4.00; it will retail for \$7.00. The figures are based upon ERNST & ERNST, *supra* note 28, which accounted for more than 84% of the sales of adult tradebooks, but lesser percentages of the sales of juvenile, religious, and professional books. The figures in the example rest on the submissions of hardbound adult tradebook publishers modified somewhat after discussion with individual publishers.

vantage by spreading fixed costs over a large number of copies.⁶⁹

The copier's cost advantage today is somewhat greater than

<i>Fixed Costs</i>	<i>First Publisher</i>	<i>Copier</i>
Plant	\$3,900	\$ 725
Editorial	1,300	450
Other overhead	2,800	2,000
Selling	650	650
Promotion	1,350	675
Total fixed cost	\$10,000	\$4,500
<i>Variable Costs</i>		
Manufacturing	\$.70 per copy	\$.70 per copy
Warehousing and shipping	.20 " "	.20 " "
Royalty	.70 " "	.00 " "
Selling	.13 " "	.13 " "
Promotion	.27 " "	.14 " "
Publishers' total variable cost	\$2.00 per copy	\$1.17 per copy
Retailing markup	\$3.00 " "	\$3.00 " "
Total variable cost	\$5.00 per copy	\$4.17 per copy

With these costs, at a retail price of \$7.00 the original publisher will just break even with a sale of 5000 copies. He may make a profit if the book sells a few more copies or if he can sell subsidiary rights. The copying publisher will not break even unless he sells 1550 copies. If he expects to sell 2500 copies, he may be able to lower his price to the retailer by \$.90 or \$1.00, producing a retail price of \$6.00 instead of \$7.00.

A tradebook copier may, however, gain an added advantage because he runs less risk that the book will prove unpopular than does its initial publisher, and risk is a more important consideration for tradebook than for textbook publishers. Yet even here most publishers interviewed seemed able to predict costs and revenues associated with a particular title with considerable accuracy. The only actual study of the matter that I could find was made by a tradebook publisher who compiled figures comparing prepublication sales estimates for fifty novels (over a ten-year period) with their actual sales. He grouped the estimates into three broad categories: a) sales of fewer than 5000 copies; b) sales of 5,000 to 20,000 copies; c) sales of more than 20,000 copies. Forty-eight of the fifty books had been placed in the right category prior to publication. Most interviewed publishers suggested that big mistakes are rare. Of course, extreme accuracy of prediction may be more important in the case of tradebook publishing than in other forms of publishing or in many other industries because of the precarious financial plight of many tradebook publishers. Some tradebook publishers said they rarely lost money on a book. Others stated that only a few books are very profitable, only a few lose heavily, and most hover near the "break even" point, often incurring small losses.

In any event, a copier must undergo some of the risks that an initial publisher bears. Tradebooks sell rapidly, producing the bulk of their sales between six and eighteen months after publication. A second publisher must enter the market quickly before it becomes saturated. Yet if he enters quickly he sacrifices ease of predictability. Moreover, books of "borderline" profitability, where risk is likely to be an important factor, are also books that do not often sell enough copies to attract a copying publisher. See p. 301 *infra*.

⁶⁹ The copier's advantage will also decline if he must bear special costs — if, for example, he wishes to reproduce the book with particular speed, or if he has difficulty finding retailers for a book that is already on the market.

copiers enjoyed in previous generations. In the nineteenth century copiers could not photograph a book's pages and thus had to bear the heavy cost of composing type for printing; nor did they benefit much from promotional expenditures, for publishers then seem to have advertised much less than now.⁷⁰ More importantly the copier's cost advantage may decline somewhat in the near future. With the aid of computers, printers may be able to make printing plates directly from a typed manuscript, eliminating the cost of retyping copy on, for example, a linotype machine.⁷¹ Computers may also lower inventory costs by making possible the printing of books "on demand."⁷²

In any event, one cannot deny the existence of a fairly large cost advantage at present.⁷³ The advantage shows that copying publishers could sell books profitably at prices that will prove unremunerative for initial publishers — a fact that may lead some to copy books soon after they appear on the market. On the other hand, the cost difference does not prove by itself that copying publishers will be able to drive a book's price low enough soon enough to prevent its initial publisher from earning a profit; nor does it show by itself that several publishers will often attempt to sell the same book.

(ii) *Countervailing Forces: The Initial Publisher's Advantages of Lead Time and the Threat of Retaliation.* — A book's initial publisher will ordinarily enjoy several advantages that may partially offset a copier's lower production costs. For one thing, his book will reach the market first. To obtain this "lead time" in the nineteenth century an American publisher would often agree to pay a popular English novelist, such as Sir Walter Scott, substantial royalties for early proofs of a new novel.⁷⁴ The

⁷⁰ See THE COST BOOKS OF CAREY & LEA, 1825-1838 (D. Kaser ed. 1963). The cost books contain only a few scattered references to advertising expenditures.

⁷¹ See LEE, *supra* note 57, at 57-61.

⁷² Yet other developments, possibly in the most distant future, may widen the cost gap. It may, for example, prove possible to store word images in inexpensive "cassettes" that can be "read" when inserted into home television sets; authors' royalties may account for an enormous share of initial "cassette" costs, providing a large cost advantage for any copier who need not pay royalties. Or books may come to be used less frequently than computers and microfilms for disseminating information, a change that would alter the publisher's role. Nonetheless, despite the possibilities of future change, the discussion will proceed primarily in terms of present technology, with the hope that the principles involved may be applied *mutatis mutandis* to take technological development into account.

⁷³ Since the size of a copier's cost advantage depends on the type of book and size of the edition, average figures are not very helpful. Yet if one wants a rough, typical, "ballpark" figure, one might think of the costs that a copier would save as varying between 25% and 30% of total production and selling costs (including retail distribution).

⁷⁴ THE COST BOOKS OF CAREY & LEA 1825-1838 at 48 (D. Kaser ed. 1963)

proofs would be rushed to America before any other publisher could compete by obtaining a copy of the book's English edition.⁷⁵ Books were sold with sufficient speed that a few days lead time could spell the difference between profit and loss.⁷⁶ In fact lead time was important enough that many English writers earned more from the sale of advance proofs to American publishers (despite lack of copyright protection in America) than from the copyright royalties on their English sales.⁷⁷

Lead time should still prove advantageous to publishers of tradebooks. Publishers expect a tradebook to begin to return a profit within a few months of publication if it is ever to do so. By the time a copier chooses a book, prints it, and distributes it to retailers, he may be six to eight weeks behind, by which time the initial publisher will have provided retailers with substantial inventories. It is unlikely that a price difference of less than a dollar will lead many retailers or customers to wait for a cheaper edition, for hardbound book customers do not seem to respond readily to price reductions. They are not willing, after all, to wait for a cheaper paperbound edition.⁷⁸ And, if the copier tries to cut his competitor's lead by rushing publication, he will incur special costs that will cut into his price advantage. In the case of textbooks, however, lead time, while helpful, should prove less important, for college texts often take a year or more before they begin to earn a profit, and elhi texts may not begin to earn a return for two or three years.

Second, the copier will be somewhat constrained by fear of retaliation by the initial publisher. Publishers in the past protected themselves from copying by producing punitive "fighting editions," which they would sell below the copier's cost.⁷⁹ Even

(\$1475 for the life of Napoleon Bonaparte). See also Testimony of J. Murray, Esq., 1878 MINUTES 62: "[Q]uite recently one house in America (and I have no doubt that it is not a solitary instance) gave as much as 1,000 [pounds] to the Livingstone family for the early sheets of his 'Last Journal...'"

⁷⁵ D. KASER, BOOK PIRATING IN TAIWAN 8-9 (1969).

⁷⁶ See D. KASER, MESSRS. CAREY & LEA OF PHILADELPHIA 91-116 (1957).

⁷⁷ Mallet 1.

⁷⁸ Most publishers interviewed seem to think that price cuts will not significantly increase the demand for a book. Yet they also believe that a price increase will significantly cut sales. This reaction may reflect a book-buying public that is reluctant to pay more than the "going rate" for a book (a rate fixed in their minds by noticing the price normally charged when a book comes out), but a public that is not particularly interested in book bargains. If so, a lead time of a few weeks should prove advantageous despite a somewhat higher price.

⁷⁹ See Plant 173. As Professor Plant notes, there was also "a kind of tacit understanding among the larger publishers in America that the books published by one should not be pirated by another." Testimony of Prof. T. Tyndall, 1878 MINUTES 316. That "tacit understanding," however, seems to have been disregarded fairly often. The threat of "retaliation" was used to try to hold copiers in line.

if the fear of antitrust action makes the modern publisher hesitate to set a price that is lower than that of the copier,⁸⁰ the antitrust laws should not prevent him from *meeting* the copier's price.⁸¹ And at equivalent prices he will benefit from marginal advantages, such as somewhat better quality reproduction,⁸² more fully-developed channels of distribution, or an ability to proclaim his the "authorized" edition.⁸³ When, for example, a copying publisher recently put out an \$8.50 edition of a book about ancient houses, the book's original publisher responded with a new edition priced at \$4.00 and, because of his better established distribution channels, drove the copier from the market.

A copying publisher, faced with the problems of "lead time" and "retaliation" is unlikely to see much profit in copying low-volume titles. It seems unlikely, for example, that a publisher thinking of copying the type of tradebook that now sells about 4000 copies, would count on selling the 2000 or more copies needed to earn a profit.⁸⁴ Nor would a publisher seem likely to compete in the production of a low-volume text as long as its initial publisher is not earning unusually high profits from its sale.⁸⁵ Indeed, pre-1958 British books, which are often in the public domain in America, are rarely copied—I should think because their market is too small to make copying attractive.

Professor Kaser tells of the "retaliation" that Carey & Lea provided Harper's when the latter pirated one of their English novels. Carey & Lea made a strenuous and successful effort to produce and to place Harper's next English novel in the New York bookstores before Harper could do so itself. "Because of their efforts Bulwer's 'Rienzi' was for sale in all the New York bookstores one day earlier than Harper's edition of the same work." See D. KASER, *MESSRS. CAREY & LEA OF PHILADELPHIA* 143-55 (1957).

⁸⁰ There is some risk that a price below average cost, but higher than variable cost, might be considered "predatory pricing" in violation of section 2 of the Sherman Act, 15 U.S.C. § 2 (1964) or section 2 of the Clayton Act as amended by the Robinson-Patman Act, 15 U.S.C. § 13 (1964). See P. AREEDA, *ANTITRUST ANALYSIS* 166-68, 647-66 (1967).

⁸¹ Note that the original publisher's variable costs *excluding* authors' royalties are not much greater than the copying publisher's costs. The author may agree to a reduced royalty on a fighting edition to retaliate against a book "pirate."

⁸² See Price, *How Publishers Are Leaving the 1950's Behind*, *TIMES LITERARY SUPPLEMENT* 821-22 (July 24, 1969).

⁸³ J. R. A. Tolkien's *Lord of the Rings* is not protected by copyright in this country. Its original publisher has reaped some advantage by proclaiming his the "authorized edition." Hurt & Schuchman, *supra* note 2, at 428-29.

⁸⁴ See note 68 *supra*.

⁸⁵ Professor Plant, after examining the 1878 MINUTES, concluded that "perhaps the most important check on the rival publisher . . . was the low-price policy which the American publishers adopted." Plant 173. As long as prices for low volume texts and tradebooks do not rise far above the costs of the initial publisher, it seems unlikely that a competitor will enter the market. At the same time the possibility of attracting new entry may have the beneficial effect of keeping prices near that level.

A copier is more likely to compete in the sale of those trade-books (variously estimated at ten to fifteen percent of total trade-books) that sell more than a few thousand copies. Yet the copier's cost advantage is smaller for such higher volume books. And with only one or a few copies we may find that price competition is not sufficiently fierce to force initial publishers to incur a loss; "duopoly" or "oligopoly" pricing may leave a profit for all publishers on books worth copying. That competition in the sale of such books is feasible is suggested by existing competition in the sale of popular government documents, such as the *Warren Report*, or of classics in the public domain, and by the fact that publishers paid popular nineteenth-century English writers substantial royalties despite the lack of copyright protection.

Nonetheless, to introduce competition and lower the return that a "best-seller" earns may make tradebook publishers less willing to run the risk of publishing books of unpredictable future popularity.⁸⁶ Moreover, copiers are likely to compete in the production of popular texts or reference books. These books, unlike "best-selling" novels, may involve heavy fixed costs which the original publisher does not expect to recoup for two or three years or more. Such books present the strongest examples for those who fear that the threat of competition will make publishers hesitate to produce books. Yet publishers will continue to produce them as long as they can earn a reasonable profit on their sale; and copyright would be unnecessary even in this category if it proved possible to assure them this profit by developing currently underutilized marketing techniques for channeling money from reader to publisher — techniques that we shall now discuss.

(iii) *Other Ways to Sustain Publishers' Revenue.* — If, in the absence of copyright protection, an initial publisher of, say, a text with a long time horizon and large initial expenses fears that copying will destroy his profits, buyers — anxious to obtain the book — may find ways of assuring him enough revenue to operate profitably. Buyers, individually or in groups, might contract to buy the book in advance of publication, before copying is possible. Or organized groups of buyers may place a large enough order for an initial printing (either before or after publication) to assure the publisher a profit. Also, the government could subsidize the publication of certain high cost books.

If many, or most, of the potential buyers for a particular type of book can be organized into a group that can negotiate with publishers, the danger that copying will make book production

⁸⁶ The number of such books is probably not large. See note 68 *supra*.

unprofitable diminishes. For if a publisher can convince the group that he needs a substantial "contribution" to cover the costs of production, the group will often prefer to pay the money than to lose the book.

A system that relies upon organizing buyers into groups, however, will have several disadvantages. First, to form a buyers' group will impose administrative costs. Yet such costs are not always unbearable; one need only think of a professional association of physicists using dues to support a physics journal,⁸⁷ of a group of school districts jointly contracting for the development of a textbook, or even of a book club that provides its members with novels.

Second, a buyers' group may require the *delegation* of the buying decision to a fairly small number of persons, which some may fear would interfere with individual choice. Yet in some areas, such as textbook buying, that decision is already delegated to a few teachers or to school administrators.

Third, one might argue that publishers will often require buyers to pay for books in advance of publication (though a publisher may also sometimes wait until *after* publication and then try to sell a large batch of books to a group).⁸⁸ And buyers may prove reluctant to buy a "pig in a poke" — to pay for a book before they can see it. I doubt, however, that this argument is of great importance. Buyers *now* pay some of the cost of producing unpopular, unsuccessful books when they buy books that are successful, at least to the extent that profits on successful books must compensate for the losses incurred on unsuccessful books if the publisher is to obtain an overall profit sufficient to stay in business. Advance payment should not increase the total price paid for the books that buyers want.⁸⁹

Fourth, "freeloading" may prevent the group from raising enough money to obtain all of the books it wants.⁹⁰ In a group where negotiation between members is permitted, each member

⁸⁷ The subscribers' dues ordinarily pay part of the publisher's costs, but those costs do not include royalties. Contributors to such journals write without monetary compensation.

⁸⁸ A publisher should ask for payment in advance because his bargaining position will be stronger. The buyers will believe that if he is not paid his costs (including a reasonable profit) he will not produce the book. If the book has already been produced, however, he may threaten the buyers' group that he will not release it unless they pay him as much, but his threat will look like a bluff. One important exception to this strategy is the case of the author who has a much higher opinion of what he is likely to produce than does the buyers' group. It may pay such an author to produce his work first and then sell it to the group.

⁸⁹ In fact, paying in advance may spread risks, thereby lowering the cost of producing a book.

⁹⁰ See, e.g., R. MUSGRAVE, *THE THEORY OF PUBLIC FINANCE* 131 (1959).

may find it in his interest to understate the value to him of the books that the group is considering buying. If Smith, for example, believes that ninety other members will each put up a dollar for a novel and that the author will create it for ninety dollars, he can claim that the book is worth nothing to him or refuse to join the group and thereby receive the book without paying a royalty to the author.

Use of this strategy may lead to a decline in book production, *but only when* understatement of preferences means that insufficient money can be raised to pay an author or publisher what he needs to create the book. In fact, were negotiations costless, one would expect to find few, if any, cases in which "freeloading" hindered book production. Each member might suspect the others of understating their true preferences and engage in a game of bluff and counter-bluff for a while. But when he sees that, as a result, the book may not be published, he will begin to express his true feelings. Each "understater," after all, would by definition rather pay more money than do without the book.

Of course, negotiation is not costless, and it takes time. Thus, the larger the group and the more difficult it is to negotiate within it, the more likely that a strategy of "understating" will hinder the production of books that the group would like. Still, even large groups can minimize the freeloading problem. Some, for example, may offer advantages that lead most potential buyers to join and pay dues rather than remain outside. Members may receive books before they are reprinted ("pirated") for others. Or certain books may be available only to organization members because it does not pay to reprint them. Further, many groups, such as universities, foundations, and school districts, may believe it unethical to become freeloaders; their rules or practices may minimize its occurrence. Finally, insofar as buyers' groups finance the creation of a book, they spread the risk of the book's failure among a large number of people. Since 1000 men are more willing to risk one dollar each than one man \$1000, risk spreading lowers risk costs. And lower risk costs should increase production, thereby offsetting the freeloading problem to some degree.⁹¹

⁹¹ To the extent that "risk savings" are large in relation to the cost of organizing the group, one should expect to find buyers' groups now in existence, for the copyright law does not prohibit them. Advance payment by government for the creation of an elhi science text, for example, may represent the risk factor at work.

A second offsetting factor may arise if some members of a buying group decide that a proposed work is particularly valuable to them and offer to contribute some extra amount, thereby encouraging the production of currently unpublished works.

Two examples — while speculative — suggest that if there were no copyright protection, buyers of textbooks could organize without great difficulty. The major buyers of elhi texts — representatives of state and local school boards — are comparatively few in number. In fact, the executive director of the textbook publishers' trade association has pointed out that "twenty-three states adopt elementary textbooks at the state level and eighteen states adopt secondary textbooks in the same manner."⁹² If publishers of high cost elhi texts can convince school districts that competition will make publication unprofitable, one would expect that a number of districts would join together and guarantee the publisher sufficient revenue rather than lose the book. Because the number of separate buyers is not large, and because they are public bodies, the freeloading problem should not prove serious.⁹³

Without copyright protection, it may also prove possible to organize college text buyers in a different, but related, way. A publisher contemplating a particularly expensive text could ask college teachers who might use it to make a *conditional* commitment: they would promise that *if* they adopt the book, the college will pay, say, an extra five dollars per copy used, for a period of, say, three years. The college would pay for the books when they are used and would recover their cost by charging students. Since publishers now send salesmen through most colleges, obtaining these pledges should involve little extra manpower. Moreover, it may prove easier under such a system (which would allow competition in the publication of individual books) than under the present copyright system (which protects the publisher from direct competition) to prevent text prices from rising far above production costs.⁹⁴ Colleges will have more bargaining power when

⁹² *Study of Instructional Technology* 8-9, Statement of A. McCaffrey to President's Commission on Educational Technology, March 1970.

⁹³ On the other hand, the very fact that school boards are public bodies may cause trouble. The boards may, for example, hesitate to discard books that they have paid for in advance, even if those books do not meet expectations. In principle, if there are benefits in maintaining a wide choice among texts, a rational buyer, paying for text creation in advance, would pay to have several texts produced and then discard those he does not want. If school boards act in this way, however, taxpayers may believe that they are acting not rationally, but wastefully. It may prove wiser politically to hide the cost of providing a wide choice by allowing some books to fail to recover their costs and then paying publishers higher prices for those books that are successful.

⁹⁴ That the prices of at least some college texts are considerably higher than their costs is suggested both by the cost and profit figures provided at p. 315 and notes 132 & 133 *infra*, and by the fact that many texts seem to increase in price as they go into subsequent printings though the cost of producing copies should fall. To take one example, J. JOHNSTON, *ECONOMETRIC METHODS* (1963) retailed for \$7.95 in 1967 and the same book without changes retailed for \$9.95 in 1969.

publishers seek advance commitments than do students when they buy assigned texts; and college officers may refuse to make advance commitments that they believe will provide the publisher with supernormal profits.⁹⁵ Competition from second or third printings of other texts should also prove more effective at keeping prices down in a world without copyright protection than in the present one.⁹⁶

Organizing buyers may prove considerably more difficult in the case of tradebooks, for the number of tradebook buyers is large and there is no existing institution that could easily serve as a focal point for organization. Of course, book clubs might sell advance subscriptions. Many such clubs now pay publishers large sums before a book is published for the right to distribute it.⁹⁷ And nearly as many hardbound and paperback tradebooks are now sold through book clubs as through other outlets.⁹⁸ Yet a large market for tradebooks will probably remain outside bookclubs, for some readers will not wish to commit themselves in any way without first seeing the book. Many others buy tradebooks as gifts or on impulse while browsing. The presence of a large, unorganized group of potential buyers suggests that the freeloading problem may prove severe and that we cannot count on buyers' groups to make up for lost publishing revenue.

In sum, without copyright protection organizing buyers to channel needed funds to publishers may sometimes prove difficult but will often prove practical. The ability to organize buyers — along with the initial publisher's "retaliatory" and "lead time" advantages — should severely *limit* — even though it may not eliminate — the revenue loss which publishers fear that unrestricted competition threatens.

In the absence of copyright protection, one might also con-

⁹⁵ When a college text publisher asks a teacher for an advance commitment, the college may require that an administrator negotiate its scope. The administrator will have bargaining power, for he will know that a refusal to agree to a publisher's request will not prevent publication as long as a publisher could produce it without any commitments (as should often prove possible) or with a smaller commitment than requested.

⁹⁶ Without copyright protection, the threat of competition should force down the price of subsequent printings of popular texts. Lower prices for such books may also force down the prices of new books. And it is unlikely that we are worse off with cheap editions of older books even when their presence makes it uneconomical to publish certain ones. *Cf.* pp. 326-27 *infra*.

⁹⁷ See Price, *How Publishers Are Leaving the 1950's Behind*, *TIMES LITERARY SUPPLEMENT* 821-22 (July 24, 1969).

⁹⁸ The 1967 CENSUS OF MANUFACTURES reports that in 1967 receipts from sales of general books (basically tradebooks) through book clubs amounted to \$155.8 million. Sales of adult tradebooks sold primarily through book sellers amounted to \$216.7 million.

sider using government subsidies to maintain publishers' and authors' revenues. At present the government subsidizes scientific writing by paying for nearly two-thirds of all research and development work done in America⁹⁹ and by paying a sizeable portion of the cost of disseminating the results.¹⁰⁰ Insofar as the government pays for a book's creation and subsidizes its dissemination, a copyright fee is not needed to assure production and to charge such a fee will restrict the book's distribution. Moreover, government subsidies can, and do, take "spillover" effects into account. The government may pay for a work containing a scientific theory, for example, that radiates benefits far beyond the circle of its readers. Because the sales revenue that such a book generates may not cover its costs, its subsidy by the government is necessary and desirable.

In principle, substituting government funds for money raised through copyright may prove economically advantageous, for it can secure production without restricting a book's dissemination. But I doubt that one can devise an administratively practical government financing system that could be *substituted* on a wide scale for copyright protection without diluting much of this theoretical economic benefit.¹⁰¹ Moreover, increased reliance upon

⁹⁹ In 1965 the federal government financed 64% of all research and development, though it performed only 15% of R&D work, while industry performed 70% and universities 12%. The federal government financed 63% of fundamental research; it performed 14% of such research, while universities performed 58%. OECD, *REVIEWS OF NATIONAL SCIENCE POLICY, UNITED STATES* 34 (1968).

¹⁰⁰ The OECD estimates that in 1967 the federal government spent nearly \$275 million on disseminating scientific and technical information. It spent nearly \$80 million directly on the publication and distribution of information in published (as opposed to typescript) form. *Id.* at 226.

¹⁰¹ One would want to replace copyright protection with a system of government financing if it had the following two economic virtues: 1) A book will be created and published if, but only if, its total social benefit (*i.e.*, its value to those who buy or use it) equals or exceeds the costs of producing it. 2) A *copy* of such a book will be made and disseminated if, but only if, the *additional* social benefits that the copy yields equal or exceed the additional cost of making and distributing it. But systems with these virtues require the government to estimate with reasonable accuracy the value of a book to every user of each copy. This fatally flaws any such system, for to estimate accurately the social value of even one book, let alone every book, is virtually impossible. And to make publication decisions turn upon rough estimates of social value made by the government invites not only censorship but corruption.

Systems of government financing that are comparatively simple to administer have fewer economic virtues. It has been suggested, for example, that the government might pay lump sums to authors for their manuscripts and then hire publishers to produce them, presumably on the basis of competitive bidding. Such a system, however, provides no safeguard against the production of books whose costs exceed their value unless we reimport into it those very estimations of a book's value that are so difficult to make and to administer. A different system might have the government pay an author a flat sum per copy sold (say, 10 cents

government financing can raise various noneconomic problems. One cannot be certain that the government would prove willing to devote any more money to book production than it does at present. It may prove particularly difficult to persuade Congress to increase appropriations to support novels, for instance, or religious biographies.¹⁰² More importantly, increased reliance upon the government for financing book production increases the risk of censorship. As Professor Baumol has pointed out, "[I]t is not easy to forget the moment when counsel for the House Un-American Activities Committee [investigating the Federal Theatre Project] asked Hallie Flannigan: 'You are quoting from this Marlowe. Is he a Communist?'" ¹⁰³ The considerations are too closely balanced to allow an assertion that government financing constitutes a satisfactory alternative to copyright protection. For this reason, I point to the possibility of government financing as a way to reduce the need for copyright protection, but I hesitate to rely upon it.¹⁰⁴

per copy). This system would prove easy to administer, and, since buyers would pay most of a book's production costs, there would be some assurance that the book was wanted. Yet this system, too, is very imprecise. In some cases it may lead to the production of copies of books that are not worth to their users what it costs to make them. In other cases, it provides no guarantee that book prices will fall; indeed, where circulation is too limited to justify more than one publisher, that publisher would presumably price his books as at present, while the author pockets both royalties and subsidy.

It is, of course, possible to devise many variations of the basic "government subsidy" theme. Rather than clutter this article with complex detail, however, I shall simply report that I have found none that combines significant economic advantage with administrative simplicity.

¹⁰² Although a large group that mildly desires a particular type of product might be effectively organized as a *market* force (as each will devote a small amount of money to securing the product), such a group cannot be easily organized as an effective *political* force (for very often no one values the product sufficiently to make his vote depend on whether he secures it). The interest in securing literary, musical, and artistic production via subsidy may fall prey to these political realities. Cf. J. BUCHANAN & G. TULLOCK, *THE CALCULUS OF CONSENT* (1965).

¹⁰³ W. BAUMOL & W. BOWEN *supra* note 26, at 374. The censorship problem can easily be overstated. Professor Baumol, in his comprehensive survey of the economics of the performing arts, suggests that in Western Europe, despite government support, the performing arts are manifestly free from political pressure or interference; and in America, despite sizable grants given by governments to encourage writing, composing, and performing, "we encountered no case of interference by state or local governments, even where the amount of assistance provided has been relatively large." *Id.* Still the danger cannot be ignored. Bringing subsidies into the open may lead taxpayers to ask, "Why is our tax money going to support this pornography (this subversive material)?" And Congress may respond with restrictions.

¹⁰⁴ The policy question that we face is whether there is some subsidy system of the sort referred to in note 101 that is better than the present law of copyright. From an economic point of view, there may be a subsidy system that, despite the

(b) How Badly Is Copyright Needed?

The discussion so far indicates that without copyright protection competition should produce no more than a modest fall in publishers' revenues, and even that decline would normally be limited to the sale of high volume works. Yet some revenue loss is inevitable. Does it provide cause for serious social concern? Answering this question, while necessary, requires considerable speculation.

In my view, a modest fall in *college text* publishing revenue should not cause much concern, for several factors minimize the extent to which any such fall could cause significant harm: If increased competition threatens the royalty income of their authors (who are primarily college teachers¹⁰⁵), they may turn from writing texts to scholarly articles or they may devote more time to teaching. In view of the spillover benefits that these other activities provide, one cannot conclude that substituting somewhat more of them for somewhat fewer college texts would prove socially harmful. Moreover, if it begins to appear to the university community that not enough texts are being created, the community could exercise certain nonmonetary pressures (suggestions from deans, professional recognition, etc.) and possibly monetary inducements (extra salary, prizes) to encourage the faculty to spend more of its research time writing texts. Colleges *now* pay for the creation of texts to some extent by paying salaries to their authors and allowing them time to write. Without copyright protection colleges and their students, considered as a group,¹⁰⁶ should be willing and able to spend more money on salaries (if necessary), for they would receive a benefit that is at least equivalent in the form of lower book prices. Finally, many college text publishers now earn profits that are

shortcomings mentioned in note 101, emerges slightly ahead of copyright. After all, the copyright system restricts distribution; it forces the expenditure of resources in securing permissions; and in its efforts to secure production it ignores spillover effects. On the other hand, from an administrative and fund-raising point of view, the present copyright system would seem superior. In any event the advantages of complete reliance upon a system of government subsidy are not clear enough to warrant such a radical departure from the status quo. This conclusion does not mean that we should abandon the use of government subsidies to *supplement* revenue raised through sales under copyright. Through use of *supplemental* subsidies we can take some spillover effects into account without creating the drawbacks of full-fledged reliance upon a subsidy system.

¹⁰⁵ See McCaffrey, *supra* note 92, at 7-8.

¹⁰⁶ If textbook authors are located in only a few universities, however, the monetary loss to the authors and the monetary gain to the students may be distributed unevenly across the country. In such a case, unless contributions are made by the "student" universities to the "author" universities, the latter will be forced to subsidize the former.

higher than the average level of profits in manufacturing. To some extent they may simply absorb any revenue fall;¹⁰⁷ and, if increased competition has produced the fall, book output may tend to expand, not contract.¹⁰⁸

Elhi texts provide greater cause for concern. Like college texts they are more likely to produce spillover benefits than many other books. But, unlike college texts, one cannot easily predict that a fall in income would lead their authors (often teams of specialists employed by publishers¹⁰⁹) to turn to fields that are of at least equivalent social value. Moreover, elhi text publishers' profits are, on average, lower than those of college text publishers;¹¹⁰ they are less likely to absorb a revenue loss. Further, unlike the successful best selling tradebook, high volume does not necessarily mean a bonanza for a single author; elhi texts are often written by teams of professional writers who are paid only moderately for work on even a very popular textbook. They are likely to prove sensitive to a cut in income. Thus a fall in the revenues generated by high-cost, high-volume elhi texts (or similar books, such as subscription reference books) may well induce a harmful loss of production. In the absence of copy-right protection one would have to rely upon buyers (such as school districts) organizing to minimize the amount of any such loss.¹¹¹

Tradebooks present something of a middle case, though any conclusion about social harm here must be still more speculative.¹¹² I have already indicated a belief that increased competition will not bring about a revenue loss sufficient to deter

¹⁰⁷ See p. 315 *infra*.

¹⁰⁸ Additionally, since the scholarly writings of one author are the raw material for another's creation, the output response to increased availability of raw material (brought about through the elimination of copyright restrictions) will counteract to some extent the response to lower payment for what is produced.

¹⁰⁹ McCaffrey, *supra* note 92, at 7.

¹¹⁰ See p. 315 *infra*.

¹¹¹ It may also prove possible for school districts to pay elhi text creators directly, as universities pay teachers who write college texts, for they may work in part for school districts.

I suspect that it would be considerably more difficult for buyers of subscription reference books to organize alternative methods for paying their producers than for buyers of elhi texts to do so. Only five percent of all subscription reference sales are to schools and libraries (buyers that might be organized with relative ease). Seventy-three percent of their sales are accounted for by home and office users. S. HUNT, 1968 ANNUAL SURVEY OF SUBSCRIPTION REFERENCE BOOK PUBLISHERS [hereinafter cited as HUNT SURVEY]. Although these latter users may be located with ease, their numbers make serious freeloading problems probable.

¹¹² We must speculate both because of the difficulty of making firm judgments about the element of risk involved in tradebook publishing, see note 68 *supra*, and because of the lack of specific information about the extent to which a decline in royalty payments to the writers of high volume tradebooks will lead them to turn to other occupations.

publication of *high-volume* tradebooks.¹¹³ I did not mention the fact that publishers will also lose revenue because of increased difficulty in selling subsidiary rights,¹¹⁴ such as movie rights or rights to paperback publication.¹¹⁵ I doubt that the additional loss will prove either terribly large¹¹⁶ or of very significant effect.¹¹⁷ Despite competition and a loss of some subsidiary right revenue, publishers still ordinarily should recover their own costs.¹¹⁸ Although the author's royalty may fall, he will still be paid — in fact often paid well.¹¹⁹ And there are at least a few general reasons for believing that authors of best sellers will not prove very sensitive to changes in revenue: these authors have incomes that are comparatively large; some have a particular talent for

¹¹³ See p. 302 *supra*.

¹¹⁴ In the case of textbook publishers, subsidiary right income is of little importance. About 1.5% of elhi textbook publishers' income and less than .5% of college textbook publishers' income arises from sources other than the sale of books. HUNT SURVEY 8, 10. However, subsidiary right income amounts to 8% of the revenue of publishers of adult hardbound tradebooks, ERNST & ERNST 10, and perhaps up to 40% of the writing income of many of their authors. LORD 12.

¹¹⁵ These rights also include translation rights and various miscellaneous rights.

¹¹⁶ The figures provided in note 114 *supra* considerably overstate what would be lost without copyright protection. One third of publishers' subsidiary right revenue, for example, comes from the sale of paperback rights. ERNST & ERNST 10a. Without copyright protection, an initial publisher can retain some (but not all) of this revenue by selling those rights prior to initial publication or by bringing out a simultaneous paperback edition himself. The fact that paperback publishers now often pay large premiums for publishing rights prior to a book's initial publication suggests that simultaneous publication is often feasible. See Price, *supra* note 97, at 821-22. Book club royalties account for another third. ERNST & ERNST 10a. Many book clubs would also pay for publication rights prior to an initial publication. In fact, the "simultaneous edition" technique should moderate the loss of subsidiary right income in many areas, ranging from films to foreign language editions.

¹¹⁷ Some subsidiary right income is highly speculative and its influence upon the decision to create or to publish must be remote. Also, much subsidiary right income is earned by the creators and publishers of successful high-volume works — works which are few in number and which may well be published without the additional incentive that subsidiary right income provides. Lord reports that in 1958 only "1 book writer in 4 drew personal income from paperbound reprints, 1 in 5 earned from bookclubs; and 1 in 15 drew income from TV rights. Motion picture rights to books are by far the most lucrative subsidiary source for book writers but less than four percent of all book writers each year realize such income." LORD 151. In light of these considerations and those of the previous note it is difficult to say that the power to sell subsidiary rights is essential to avoid a serious production loss.

¹¹⁸ See p. 302 *supra*.

¹¹⁹ If, as appears to be the case, tradebook publishing is highly competitive, a decline in publishing revenue will be pushed back to the author in the form of a decline in royalties. The royalties that American publishers paid popular British authors at a time when they were not protected by copyright, however, were still considerable. T.H. Farrar testified in 1878 that "I have been told . . . that the practice of some of the great houses is to pay 10 percent., upon the price of copies sold in America." 1878 MINUTES 268.

writing; and at least a few may be more interested in literature than money. Thus, many may continue to write despite some loss of income.

A more likely victim of a fall in revenue is the tradebook of unpredictable popularity, perhaps an author's first novel. Since competition will diminish the size of the best seller's payoff, it will make publication a greater gamble. Publishers may then prove less willing to finance the production of such a book and authors will have to bear more of the financial risks of initial publication.¹²⁰ Some authors may then prove either unwilling or unable to continue writing. My discussions with publishers, however, lead me to think that the number of such unpredictable manuscripts is not large.¹²¹ And one cannot conclude that society is much worse off if some writers, discouraged by longer odds, take up some other occupation.¹²² Many other jobs open to the potential author also involve risk and innovation without protection from competition or copying,¹²³ and one simply cannot predict whether the marginal writer will produce a more or a less valuable social product in one of these other fields.¹²⁴

¹²⁰ Cf. note 68 *supra*.

¹²¹ The relevant sort of unpredictability is uncertainty as to whether a book will lose money or become a bestseller (*i.e.*, sell enough copies to attract competition). This degree of uncertainty would seem unusual, even in tradebook publishing. See note 68 *supra*.

¹²² In fact, Schopenhauer believed that "The reason why Literature is in such a bad plight nowadays is simply and solely that people write books to make money." F. SCHOPENHAUER, *On Authorship, The Art of Literature*, in *ESSAYS: FROM THE PAREGIA AND PARALIPOMENA* 5, 6 (Allen & Unwin ed. 1951).

¹²³ Consider, for example, the businessman trying a new marketing device or expanding into a new town; or the skillful doctor or lawyer who, until his reputation develops, must sell his work at a price well below its value to its recipients. Cf. Arrow, *Uncertainty and the Welfare Economics of Medical Care*, 53 AM. ECON. REV. 941 (1963). Doctors and lawyers, unlike the author (whose payment depends upon the number of copies sold), are unlikely to receive additional payment if a bit of work proves particularly skillful. Although as their reputations grow they can charge higher fees, the arguments in briefs and new methods of treatment can be copied freely.

¹²⁴ In a perfectly competitive economy without other externalities, one could not make this statement. Under such conditions (leaving aside certain technical economic problems) the optimum amount of writing production and dissemination would be that generated by the application of the following rule: "A writing will be produced if and only if we could pay the cost of its creation and distribution with a sum of money representing the total value of the writing to all of its users." Since readers would pay more for the work of this marginal author than he would earn in other fields, for him to leave writing would be harmful.

But the real world is not perfectly competitive. The theory of the "second best" tells us that the presence of monopoly in some sectors of the economy makes it uncertain what prices should be charged in other sectors. See Lipsey & Lancaster, *The General Theory of the Second Best*, 24 REV. ECON. STUDIES 11 (1956). Moreover, the real world is filled with externalities and risks of various sorts. Their pre-

In sum, some high cost, high volume books, such as certain elhi texts, might be significantly affected. But otherwise, on the facts available, it seems that a moderate decline in publishers' revenues would probably not produce a serious loss of production. Of course, to avoid any loss at all we need only retain and expand copyright, but that, too, has its social costs, and it is to these that we now turn.

2. *What Would Be Gained by Abolishing Copyright in Books?*

(a) *Lower Prices and Wider Distribution.* — Removing copyright protection should induce competition in the production and sale of relatively high-volume titles. Readers of these books should benefit from lower prices. But, of course, readers would be worse off if competitive prices were so low as to discourage author and publisher from producing the book. In the nineteenth century, opponents of copyright protection pointed to the enormous difference in the price of an English novel in England (where protection existed) and of the same novel in America (where it was not protected) as evidence of the harm that copyright caused the reading public.¹²⁵ Yet it may have been British protection that guaranteed author and publisher sufficient income to make them relatively indifferent to American prices. While it is no easier today than in the nineteenth century to determine precisely the *extent* to which abolition would result in lower prices, there is reason to believe that without protection the price of both tradebooks and textbooks would drop, though textbooks by a more significant amount.

(i) *Tradebooks.* — The prices of hardbound best selling novels (and other tradebooks with wide circulation) are those most likely to fall with the abolition of copyright in tradebooks. Even though the cost of producing a copy of a book declines radically as its circulation increases, hardbound editions of widely circulating bestsellers presently are priced no lower than those of other tradebooks. It is not until a paperback edition appears

sence in other occupations to which the author may turn makes it uncertain whether his giving up writing would provide a more or less socially valuable product. Finally, the fact that a writer protected by copyright can earn more money writing than in another occupation may reflect the fact that tradebook publishers can discriminate in price to some extent while other employers cannot — a fact unrelated to the social value of the writer's work in the two fields.

¹²⁵ The novel *Middlemarch*, for example, sold for £2.2s (\$10.50) in England, while in America it sold for 10s, 9d (\$2.70) in an expensive edition and 5s, 4½d (\$1.35) in a cheaper edition. 1878 MINUTES 376.

sometime after a book's initial publication that the cost benefits of wide circulation are passed on to the reader in the form of lower prices. Competition would thus force down the price of bestsellers, while authors and publishers who find the small number of bestsellers they produce very profitable¹²⁶ would continue to produce them.¹²⁷ On the other hand, the prices of tradebooks that sell only a few thousand copies — the vast majority of all tradebooks produced — would probably not fall substantially with copyright's abolition, for it is unlikely that more than one publisher could produce such a book at a profit; thus other publishers are unlikely to compete in its sale, even if they are legally free to do so.¹²⁸

Lower hardbound bestseller prices, while beneficial to their readers, are unlikely to increase a book's circulation significantly, for most bestsellers now appear in a low-priced paperback edition some time after their initial publication. It seems unlikely that increased competition can drive the price of most paperbacks down much further; in fact, since binding a book in a paper cover does not significantly change the cost of making it,¹²⁹ lower paperback prices now probably reflect strong competition between different books or an attempt by the publisher to tap a large market for low-priced books.¹³⁰ Increased competition to sell copies of the same title may allow some readers who now wait for the paperback edition to buy the book somewhat sooner. Yet the effect of subsequently lower hardbound prices in transferring wealth from author or publisher to reader is likely to be more important than its effect on book production or distribution.

(ii) *Textbooks*. — To remove copyright protection is likely to lead to a more significant fall in the price of textbooks than of tradebooks. Existing competition between textbook titles now

¹²⁶ If tradebook publishing is highly competitive, the author can obtain much of this profit by bargaining with competing publishers for the right to publish his manuscript.

¹²⁷ See pp. 301-02 *supra*.

¹²⁸ If publishers threaten to compete in the production of such a book, it is possible that the book will not be produced at all. See pp. 300-02 *supra*.

¹²⁹ See, e.g., M. LEE, *supra* note 57, at 170-71; S. UNWIN, *THE TRUTH ABOUT PUBLISHING* 153-54 (7th ed. 1960).

¹³⁰ By means of a later paperback edition a publisher can discriminate in price among customers for a single work, charging a high price to those anxious to have a hardbound edition quickly and a low price to those willing to wait. This discrimination allows the publisher to sell books at low prices to those unwilling to pay more. There is evidence that English publishers behaved similarly in the nineteenth century, putting out a later cheap edition at prices as low as those in America (where their books sold without copyright protection). The cheap edition, however, seems not to have appeared until several years after a book's initial publication. 1878 MINUTES 377-87.

seems considerably less effective at holding down prices than competition between novels and other general works. For one thing, popular textbooks appear in low-priced paperback editions with much less frequency.¹³¹ For another, textbook publishers' profits are considerably higher on average than those of trade-book publishers.¹³² In fact, average college textbook publishers' profits are at least fifty percent greater than average profits in manufacturing.¹³³ These facts about the industry may be explained in part by noting that many texts are adopted by teachers who are, within limits, fairly insensitive to price — teachers primarily interested in subject matter who do not pay for the text themselves.¹³⁴ These facts also suggest that there is considerable room for competition in the production and sale of an individual textbook title to force down prices without foregoing production — at least in the case of college textbooks.

Lower textbooks prices have particular social value. Even a small increase in the dissemination of a text or scholarly work can yield benefits the value of which far exceeds the cost of the extra books.¹³⁵ To place books in the hands of more students means greater intellectual stimulation, greater productivity, and

¹³¹ Yet the cost of producing a copy of a textbook declines rapidly as output increases, just as in the case of tradebooks.

¹³² The ERNST & ERNST SURVEY of tradebook publishers accounting for 84% of all adult tradebook revenue reports 1968 pretax book profits as a percentage of sales for adult tradebooks at 7.6%. The HUNT SURVEY of textbook publishers reports 1968 pretax book profits as a percentage of sales at 12.4% for elhi text publishers and 19.7% for college text publishers.

¹³³ One might roughly translate the college text profit figure into profits as a percent of equity by assuming that one dollar of equity generates two dollars of sales (an examination of income statements and balance sheets of leading publishers suggests that such a translation is reasonable). If we also assume taxes of about 50%, after tax profits for college text publishers as a percent of equity in 1968 were nearly 20%. The average comparable figure for all nondurable manufacturing in that year was 11.9%. U.S. FEDERAL TRADE COMMISSION REPORT, in COUNCIL OF ECONOMIC ADVISERS, ECONOMIC REPORT OF THE PRESIDENT, 1970, at 263. But, whether or not one accepts as valid this rough conversion, one cannot deny that profits as a percent of sales are far greater in college textbook publishing than in most other industries. As mentioned in note 132 *supra*, pretax profits as a percent of sales in college text publishing were 19.7% in 1968. The corresponding figure for nondurable manufacturing as a whole was 8.4% and for adult tradebook publishing was 7.6%. ERNST & ERNST 33.

¹³⁴ A study of book publishing prepared by a leading investment firm concludes that "Textbooks are sold to local and state Boards of Education on the basis of quality and treatment of subject matter rather than on price." LOEB, RHOADES & CO., BOOK PUBLISHING AND FEDERAL AID TO EDUCATION 8 (1965).

¹³⁵ Increased circulation of information would seem to be a critical factor in the creation of new ideas. Cf. Arrow, *Economic Welfare and the Allocation of Resources for Invention*, in NATIONAL BUREAU OF ECONOMIC RESEARCH, THE RATE AND DIRECTION OF INVENTIVE ACTIVITY 609 (1962).

increased research.¹³⁶ Moreover, lower textbook prices, by leaving students, universities, or school boards with more money to spend, can lead to more expenditure upon research and education — a desirable result. Thus, I suspect that price reductions are likely to be both larger in amount and socially more valuable in the case of textbooks than of tradebooks. But in both cases copyright abolition would promote more active competition, with lower prices and somewhat wider circulation, at least for the more popular books.

(b) *Eliminating the Transaction Cost of Obtaining Permission to Reproduce.* — A second burden that a copyright system can impose upon both readers and producers consists of high "transaction costs" — the administrative costs incurred in obtaining permission to copy a work. These costs are not high where there is an organized market in copies: A man who wants a whole book can buy it easily at a bookstore. Nor are they high where the permissions' requirement is not enforced against the copier, as is now true of much xeroxing. If, however, as some would urge, copyright law is to be strictly interpreted and strongly enforced,¹³⁷ transaction costs will become significant. A law-abiding user wishing to copy only a portion of a book or article, even though willing to pay the copyright owner enough money to secure his approval, will have to buy a whole book at a store or face the difficulty and cost of contacting the copyright owner, bargaining with him, and arranging for payment. To force him to buy the whole book may be wasteful, for he wishes only a part. To face him with the possibility of high "transaction costs" will also prove wasteful, particularly if it discourages him from xeroxing, for xeroxing may represent the cheapest way (in terms of actual resources expended) of obtaining a copy of what he wants.

The only extensive quantitative study of the problem to date indicates that transaction costs of this sort could prove very large indeed. The Committee to Investigate Copyright Problems estimated in 1967 that users would xerox (or similarly reproduce) more than 1.8 billion pages of copyrighted material in 1969.¹³⁸ The average length of a copied article or text selection

¹³⁶ The large subsidies that government provides research and education suggest a belief that these activities produce large spillover benefits. Discussions of the "technology gap" between Europe and America also often view education as a critical factor influencing productivity.

¹³⁷ See generally G. GIPE, *NEARER TO THE DUST* (1967); *REPROGRAPHY AND COPYRIGHT LAW* (L. Hattery & G. Bush eds. 1964).

¹³⁸ COMMITTEE TO INVESTIGATE COPYRIGHT PROBLEMS AFFECTING COMMUNICATION IN SCIENCE AND EDUCATION, *THE DETERMINATION OF LEGAL FACTS AND ECONOMIC GUIDEPOSTS WITH RESPECT TO THE DISSEMINATION OF SCIENTIFIC AND*

is slightly more than eleven pages.¹³⁹ In the unlikely event that copiers had to obtain permission¹⁴⁰ to copy each of these selections *separately*, doing so might cost \$300 million;¹⁴¹ but, even if, as is likely, arranging for permission in batches reduced these costs by ninety-five percent or more, copiers would still face transactions costs of \$10 to \$20 million, a large cost when one realizes that total royalty payments to authors in 1967 amounted only to between \$200 and \$300 million.¹⁴²

We also know that the cost of obtaining permissions in other fields is high. Music copyright clearing houses, for example, spent about \$13 million in 1964 to collect and distribute about \$57 million.¹⁴³ National educational television broadcasters estimate that the administrative costs of obtaining copyright permissions for educational television and radio broadcasts (if required under proposed legislation) would equal, on average, \$33,000 per station,¹⁴⁴ a sum equal to thirteen percent of the average station's total budget,¹⁴⁵ and greater than the total budget of many educational radio stations.¹⁴⁶

Moreover, one must remember that the total cost of imposing

EDUCATIONAL INFORMATION AS IT IS AFFECTED BY COPYRIGHT — A STATUS REPORT 71 (Office of Education, U.S. Dep't of H.E.W., Project No. 70793, 1967) [hereinafter cited as CICP]. The figures in this report were collected from a study of sixty-six key libraries, six of which were studied in detail. The figures are rough but represent the best information presently available.

¹³⁹ The "mean number of exposures per title copied" (which presumably means the number of pages copied in each selection) is 9.2 for articles and 25.8 for books. The number of articles copied is ten times the number of book selections copied. *Id.* at 60.

¹⁴⁰ Existing copyright law may allow the copying of some of these selections without permission under the doctrine of "fair use." See p. 330 *infra*.

¹⁴¹ If we assume that obtaining permission requires writing two letters (one to request, and one to grant, permission), the cost of stamps, paper, and other office supplies will equal 25 cents. If we further assume that the letters are written by secretaries each paid \$3.00 per hour and that each takes 7.5 minutes to write and to post, the permission will cost \$.75 worth of secretarial time. If the time needed to obtain permission averages three days and if the copier would be willing to pay at least \$1 to reduce this three day waiting time to zero, the total cost of obtaining one permission would equal \$2.00. The cost of obtaining 160 million permissions (the number of separate articles and book selections copied) would equal \$320 million.

¹⁴² See note 28 *supra*.

¹⁴³ See *Another Music War Threatening?*, BROADCASTING 146, 147, 152 (March 15, 1965).

¹⁴⁴ *Hearings on S. 597 (Copyright Law Revision) Before the Subcomm. on Patents, Trademarks, and Copyrights of the Senate Comm. on the Judiciary*, 90th Cong., 1st Sess., pt. 3, at 1004-05, 1021 (1967) [hereinafter referred to as 1967 Senate Hearings]. The stations estimated that they would have to pay the copyright owners an additional \$19,800 per station to obtain the clearances. *Id.*

¹⁴⁵ *Id.*

¹⁴⁶ *Id.* pt. 4, at 1298.

a permissions' requirement includes not only these administrative costs, but also the harm resulting because some potential users, willing to pay royalties but unwilling to pay the transactions' costs, will not obtain the copyrighted item. Much current xeroxing is of texts and other scientific, professional, or technical books and articles. The widespread dissemination of these kinds of works has various spillover benefits similar to those provided by the original distribution of the works copied; to discourage them from being copied is therefore particularly undesirable.

Since a strict permissions requirement is not now often enforced against xeroxing copiers, the major harms that the law currently creates in this area flow from the uncertainty and unease of those who fear that they are violating the law. But the presence of court suits and legislative proposals suggest the current situation is unstable.¹⁴⁷ The merits of those proposals will be discussed further in Part III.¹⁴⁸

(c) *Limiting the Market Power of Publishers.* — Copyright may also injure the public by allowing publishers selling *different* books to restrict competition within the industry. Although a copyright restricts competition only between copies of the same title and does not legally interfere with competition between titles,¹⁴⁹ the power to *accumulate* these exclusive licenses to publish may nonetheless, at least in theory, inhibit such competition. This power may, for example, allow a few publishers to build "stables" of popular writers. The presence of such a "stable" may strongly attract other writers with valuable manuscripts and make it more difficult for a newcomer to obtain the promising manuscripts necessary to become established in the publishing industry. If, as a result, new entry becomes difficult, well-established publishers may find that they have obtained the power to raise their prices and to resist authors' demand for higher royalties. Any such power can curtail book circulation (by raising

¹⁴⁷ See p. 330 *infra*.

¹⁴⁸ See pp. 329-38 *infra*.

¹⁴⁹ Copyright thus poses a far less serious threat to competition than patents. While a copyright restricts only the use of an author's "expression," a patent provides its owner with a monopoly of an "idea" — a fact that limits the marketing of competing products that might hold down the patented product's price. Of course, patents are more difficult to obtain than copyrights, requiring "novelty," not simply "originality." See NIMMER § 10. Still, because of their widespread commercial application, the power to accumulate patents is likely to prove more harmful to competition than the power to accumulate copyrights. Cf. P. AREEDA, ANTITRUST ANALYSIS 376-87 (1967); F. MACHLUP, AN ECONOMIC REVIEW OF THE PATENT SYSTEM (Senate Judiciary Comm., Subcomm. on Patents, Trademarks, and Copyright, Study No. 15, 85th Cong., 2d Sess. (1958)).

prices) and may even limit the number of titles produced (by restricting royalties).

The high profits that college text publishers report suggest that an "entry barrier" problem may exist in that sector of the publishing industry. Profits that are persistently higher than average, unexplained in other ways,¹⁵⁰ are an indication of market power. College textbook publishing also seems to be more heavily concentrated in the hands of a fairly small number of firms than other sorts of publishing.¹⁵¹ One must wonder why more firms have not been attracted into the industry. In another context it has been suggested that the power of the television networks to obtain exclusive licenses to copyrighted programs has made it more difficult for potential rivals to obtain the programming needed to enter that industry.¹⁵² While no firm conclusions can be drawn about either industry without a more thorough investigation, college text publishing would seem less competitive than tradebook publishing, where average profits are low and — at least according to publishers — entry is easy.¹⁵³

Whether or not accumulation of copyrights seriously impedes competition, one might stimulate competition between titles by removing or limiting by law the author's power to grant an exclusive license.¹⁵⁴ Any such limitation would, of course, lessen the protection that the copyright law now affords the publisher, and publishers would loudly protest that without exclusivity they would not be able to recoup the considerable investment needed to bring a manuscript to market. However, conceding for the

¹⁵⁰ High profits may, for example, signal an increase in the demand for a product. If so, new firms should be attracted into an industry, and profits should then fall.

¹⁵¹ The HUNT SURVEY indicates that in 1968 12 large college text publishers, each with sales of more than \$6 million, accounted for 81% of all college text sales reported (and thus accounted for more than 70% of all college text sales). The Census Bureau has reported that in 1963 the eight largest textbook publishers (not distinguishing between college and elhi texts), accounted for 54% of all sales (thus suggesting that elhi text publishing is less concentrated than college text publishing), and the eight largest general book publishers accounted for 46% of the market. BUREAU OF THE CENSUS, CONCENTRATION RATIOS IN MANUFACTURING INDUSTRY: 1963, Report prepared for Senate Subcomm. on Antitrust and Monopoly, Senate Comm. on the Judiciary, 90th Cong., 1st Sess., pt. I, 173-74 (1967).

¹⁵² See Ross & Chazen, *Federal Regulation of Cable Television: The Visible Hand*, 83 HARV. L. REV. 1820, 1833-34 (1970).

¹⁵³ See, e.g., Lacy, *The Economics of Publishing, or Adam Smith and Literature*, 92 DAEDALUS 42, 44-45 (1963).

¹⁵⁴ One possible way of achieving such a limitation would be through extension of the antitrust laws. There seems no enormous difference in principle between the application of the antitrust laws to exclusive dealerships and similar arrangements in other fields and exclusive licenses to publish. One might approach both by asking whether the arrangement is more restrictive than necessary to achieve any legitimate business objective.

moment their need to protect this investment, exclusivity is not the only way to do so. They might, for example, agree with the author to a "passover" provision that allows the author to license a second publisher to produce his book provided that the second publisher reimburses the first publisher for part of his initial publication costs.¹⁵⁵ In any event, the period of exclusivity might be limited only to the time necessary for the initial publisher to recover those expenses that a second publisher need not bear.¹⁵⁶

Limiting exclusive licenses would not necessarily injure authors. Competition in producing and selling an author's book should increase its circulation. And, as long as authors are paid on the basis of a royalty, increased circulation should mean increased revenue.¹⁵⁷ Why, then, do authors not insist upon a non-exclusive license? Many may be unaware of its advantages; others may not have enough bargaining power to insist upon any such arrangement.¹⁵⁸ It is also possible that publishers possessing market power pass on to authors some of the excess profits made by restricting entry into publishing. If so, they could truthfully tell authors that returns would fall without exclusivity. Yet no principle of copyright protection entitles either author or pub-

¹⁵⁵ Compare the possibility of profit passovers by one dealer to another to compensate the latter for advertising expenditures that work to the benefit of the former. *Cf. White Motor Co. v. United States*, 372 U.S. 253, 270-71 (1963) (Brennan, J., concurring).

¹⁵⁶ Such a time period should be considerably shorter than the 56 years of copyright protection now provided. *See* p. 325 *infra*.

¹⁵⁷ Professor Plant explains why this is so:

Where the publisher is the entrepreneur, he is concerned to maximise the surplus of aggregate receipts over aggregate costs. The author on the other hand, if paid a fixed sum per copy or a percentage of the published price, has no concern with costs. If paid a fixed sum per copy, the author's receipts will be greater, the lower the final price and the greater the number of copies sold; if he receives a percentage of the published price, his receipts will be greatest when the *gross* receipts from sales are maximised, not (like the publisher) when *net* receipts after deduction of costs are greatest. Since every additional copy printed adds something to the *aggregate* costs of an edition, *net* receipts and the publisher's profit are maximised at a smaller output and higher price than would result in the greatest *gross* receipts and author's income.

Plant 185-86.

Plant's theorizing finds support in the fact that many authors feel that their publishers do not advertise sufficiently or that they charge too high a price for the book. *See* Horvitz, *The Pricing of Textbooks and the Remuneration of Authors*, 56 AM. ECON. REV., May, 1966, at 412, 419-20 (1965 Papers and Proceedings of the Am. Econ. Ass'n).

The divergence in interest between author and publisher would disappear if they entered into a profit sharing arrangement, for then it would be in the interest of both to maximize their *joint* net profit. Yet profit sharing arrangements are unpopular, for they invite controversy over accounting methods and lead to expensive supervision of book production. *See* S. UNWIN, *supra* note 129, at 63-66.

¹⁵⁸ This fact suggests that competition in publishing is not as intense as it might be.

lisher to any return flowing from restricting competition *between* different titles. To limit the use of exclusive licenses, then, should facilitate competition without giving either publisher or author just cause for complaint.

C. Interim Conclusions

The preceding analysis suggests that the case for copyright in textbooks and scholarly works is weaker than that for copyright in tradebooks. To abolish copyright protection would bring more benefit to the textbook than to the tradebook reader: It is more probable that textbook prices now exceed the level needed to induce production; a "permissions" requirement is more likely in the case of textbooks to impose serious transactions costs upon the reader, for readers wish to copy textbooks more often than novels; and there is a much greater possibility that copyright protection helps to restrict competition *between* textbooks than *between* tradebooks. Moreover, to abolish copyright protection would threaten tradebook production at least as much as, and perhaps more than, text production. Although tradebook publishers often have leadtime advantages that textbook publishers do not share, tradebook buyers would find it more difficult to organize alternative methods for compensating publishers and authors if needed to avert a fall in production. And a fall in subsidiary right revenue is more significant in the case of tradebooks than texts. Finally, insofar as the copyright question becomes one of simply shifting income between buyers and sellers, one is more likely to be neutral (or to favor sellers) when the buyers are readers of novels than when they are students, school districts, or colleges.

The preceding analysis also indicates that the case for copyright in books considered as a whole is weak. It suggests that to abolish protection would not produce a very large or a very harmful decline in most kinds of book production. And abolition should benefit some readers by producing lower prices, eliminating the cost of securing permission to copy, and increasing the circulation of the vast majority of books that would continue to be produced. Of course, it is difficult to consider copyright "as a whole," for the need for protection varies with the type of book. Without protection the risk of loss of production is greatest among high cost, high volume books the buyers of which could not easily organize an alternative method of payment (such as subscription reference books), while buyers of college textbooks may have the most to gain.

This analysis supports those who do not wish to expand copyright protection as the proposed Revision Bill would do. But for several reasons it does not demonstrate a need for drastic revi-

sion in present law. First, many of its judgments are necessarily impressionistic, unsupported by confirming empirical evidence. One cannot be certain, after all, that most authors of bestselling novels would continue to write despite a modest cut in their incomes. Nor can one be certain about how easy it will be for buyers to organize alternative methods for paying publishers or authors if necessary to prevent a production loss. Such judgments are put forward tentatively, subject to revision as more information is gathered. Second, much of the harm that copyright threatens may be mitigated without radical change in the law. For example, the law has not impeded xeroxing by imposing a high administrative "permissions" cost because so far it has not been enforced against those who xerox. Third, one must hesitate to abolish protection without some safeguard for the high cost, high volume subscription reference book. Yet any system of protection that attempts to distinguish among different types of books may create legal classifications that are difficult for courts to administer. Fourth, expectations of authors and publishers have been built upon the present system. The "demoralization" costs of undermining these expectations may be considerable.¹⁵⁹

In sum, it is difficult to do other than take an ambivalent position on the question of whether current copyright protection — considered as a whole — is justified. One might compare this position with that of Professor Machlup, who, after studying the patent system, concluded, "None of the empirical evidence at our disposal and none of the theoretical arguments presented either confirms or confutes the belief that the patent system has promoted the progress of the technical arts and the productivity of the economy."¹⁶⁰ The position suggests that the case for copyright in books rests not upon proven need, but rather upon uncertainty as to what would happen if protection were removed. One may suspect that the risk of harm is small, but the world without copyright is nonetheless an "undiscover'd country" which "puzzles the will, / And makes us rather bear those ills we have / Than fly to others that we know not of" ¹⁶¹

One can be less ambivalent when considering incremental change in the law. The fact that the basic case for copyright protection of books is weak suggests that a heavy burden of

¹⁵⁹ See Michelman, *supra* note 29, at 1214.

¹⁶⁰ F. MACHLUP, *supra* note 149, at 79.

¹⁶¹ W. SHAKESPEARE, *HAMLET*, act III, sc. i, lines 79-82, (Variorum 13th ed. 1905). Still, our analysis shows that a simple difference between the initial costs of a publisher or author and the costs of a copier does not in itself prove that copyright protection is desirable. Rather, to point to such a cost difference can only begin, not end, an inquiry as to the need for protection.

persuasion should be placed upon those who would extend such protection and that we might look favorably upon proposals to reduce it. The matter of such incremental changes will be considered further in Part III.

III. THE DOUBTFUL CASE FOR EXTENDING PROTECTION: SEVERAL EXAMPLES

We shall attempt here to apply Part II's methods and conclusions to several current proposals that would extend the scope of copyright protection: The first would lengthen the protection period; another would bring the xeroxed copy more effectively within copyright's confines; still another would regulate computer use of copyrighted material and grant copyright protection to computer programs. It should be noted that committees of Congress have recently considered these and other modifications of existing law at length, and that the Copyright Revision Bill currently before the Senate Committee on the Judiciary contains numerous other provisions that would extend protection.¹⁶² Those here discussed are not necessarily more important than others embodied in the bill. They have been selected simply to show that further analysis of at least some of the many recent proposals for change is both helpful and necessary.

A. Length

Despite Macaulay's warning that copyright protection "ought not to last a day longer than is necessary for the purpose of securing the good [of increased production],"¹⁶³ the period of protection has grown steadily over the centuries from two years, to fourteen, to twenty-one, to twenty-eight, to fifty-six.¹⁶⁴ The Revision Bill would further extend its term to "life of the author plus fifty years,"¹⁶⁵ a period that the Register of Copyrights states would amount to about seventy-six years on average.¹⁶⁶

¹⁶² See note 10 *supra*.

¹⁶³ MACAULAY 23.

¹⁶⁴ See DOCK 65; L. PATTERSON, *supra* note 42, at 143, 183, 185, 197, 201.

¹⁶⁵ S. 543, 91st Cong., 1st Sess. § 302 (1969).

¹⁶⁶ See REGISTER OF COPYRIGHTS, COPYRIGHT LAW REVISION, REPORT TO THE HOUSE COMM. ON THE JUDICIARY, 87th Cong., 1st Sess. 50 (Comm. Print 1961). The figures provided suggest that the new term would vary between 70 and 76 years, depending upon the extent of the general increase in life expectancy. But see Guinan, *Duration of Copyright*, in COPYRIGHT SOC'Y OF THE U.S.A., STUDIES ON COPYRIGHT 473, 494 (Arthur Fisher mem. ed. 1963) ("life plus fifty years" on average will amount to a term of more than 86 years). This time period for protection applies also to unpublished manuscripts and to that degree will limit the hitherto perpetual common law prepublication copyright. See 17 U.S.C. § 2 (1964).

This continual expansion is not surprising. Holders of copyrights about to expire have a financial interest in urging extension. Authors and publishers can lead a legislature to focus on the production and "moral" arguments for protection, while no single interest group is sufficiently affected to focus legislative attention upon the problems of dissemination. An examination of the question, however, suggests that, even if the moral argument is given its due, which is little,¹⁶⁷ extension is not justified.

The "incentive" or "production" argument for extending the copyright term from fifty-six years to "life plus fifty years" is unconvincing. It must rest primarily upon the case of the "child-caring father" — the man who writes in part to provide a patrimony for his children. It is, of course, conceivable that some prospective Miltons have given up writing after learning that Milton's daughter was destitute, but it is most unlikely. More probably authors, like others, discount the value of future income, and, when discounted, the present value of a future copyright advantage is small. If, for example, an author *knew* that his book would sell continuously¹⁶⁸ for seventy-six years, an increase from fifty-six to seventy-six years of protection (an increase of more than one-third) would lead to an increase in the present value of a manuscript of only 3.6 percent (assuming a discount rate of five percent). More important, copyright office figures show that only one book in a hundred is in print after fifty-six years.¹⁶⁹ And doubtless many of these sell only a few copies. Thus, unless the prospective author has some special reason for believing that his work will survive, to increase protection by twenty years would increase the present value of his manuscript by less than four one-hundredths of one percent, hardly enough to affect his decision to write in the first place.¹⁷⁰

This fact has little to do with the economics of publishing, however. The protection period for copyrights owned by corporations has been set at 75 years in the Bill. S.543, 91st Cong., 1st Sess. § 302(c) (1969).

¹⁶⁷ See pp. 284-91 *supra*.

¹⁶⁸ That is to say, assume that the book would earn the same revenue continuously each year for 76 years.

¹⁶⁹ A comparison of copyrights taken out in 1927 and renewals obtained in 1954 reveals that 4.1% of copyrights in books and pamphlets were renewed. Guinan, *supra* note 166, at 500. The Register of Copyrights reported in 1962 that of the 1,340 books whose copyrights were renewed in 1936, 28.5% were still in print. *Extending the Duration of Copyright Protection in Certain Cases, Hearing Before Subcomm. No. 3 of the House Comm. on the Judiciary*, 87th Cong., 2d Sess., ser. 27, 60 (1962).

¹⁷⁰ Nor can one be certain just how the prospect of a revenue increase affects the production of the writer who hopes to produce a "classic" (a book still in demand after fifty-six years). As Professor Plant has suggested, extending the period of protection may lessen his incentive to produce by leading him to rest upon his laurels rather than undertake a new effort. A. PLANT, *THE NEW COM-*

Nor would extension provide significant additional incentive for publishers, for they now normally base their publication decision upon an expectation that a book will earn a return within two years (tradebooks), five years (some texts), or at most ten or twenty years (certain reference books).¹⁷¹ Moreover, many texts and reference books can be, and are, revised every few years, allowing the publisher to obtain a new copyright. (In fact, some states will not adopt a text with a copyright more than five years old.¹⁷²) Finally, publishers indicated in conversation that the presence of a copyright does not significantly affect whether or not a "classic" is republished. The fact that a copyright may make its owner more likely to republish tends to be offset by the fact that other publishers will be deterred from republishing a classic by the need to negotiate with, and to secure permission from, the copyright owner.

Any additional production inducement that extension might bring is greatly outweighed by the additional harm it will cause. First, as books become classics competition between books becomes less effective at holding down prices. A publisher with exclusive rights to Shakespeare, for example, would find it much easier to raise his price above his costs than a publisher with exclusive rights to an ordinary novel. It is not surprising that among the more valuable of the exclusive printing rights that the King granted in the sixteenth and seventeenth centuries were those to print the Bible, the Book of Common Prayer, and various literary classics—not contemporary books.¹⁷³ To take a more recent example, the popularity of *The Forsythe Saga* on television has led to a large and unforeseeable demand for copies of Galsworthy's books. The copyright owner will probably find that he can charge higher than a competitive price for those books, restricting their distribution, even though the prospect of *this* monopoly return, unlike the one that Galsworthy may have earned in 1920, is unlikely to have influenced Galsworthy himself. Because books that last beyond fifty-six years are more likely to be uniquely valuable, copyright extension makes it progressively more likely that owners can raise their prices above

MERCE IN IDEAS AND INTELLECTUAL PROPERTY 12 (1953). But see Chafee, *Reflections on the Law of Copyright*, 45 COLUM. L. REV. 503, 508 (1945).

¹⁷¹ These figures are impressionistic and derived from conversation with publishers. They are bolstered to some extent by the fact that publishers have argued that they need five to twenty years of protection when they publish works that have been produced under government contract. See Statement of Curtis G. Benjamin, Chairman of the Board, McGraw-Hill Book Company, *Copyright or Public Domain* (1966).

¹⁷² Telephone conversation with A. McCaffrey, American Book Publishers Council, Oct. 30, 1970.

¹⁷³ See L. PATTERSON, *supra* note 42, at 88-90.

the competitive level, while at the same time progressively less likely that any such power will have influenced the decision to write.

Second, as time passes persons wishing to reproduce old articles, books, designs, or other writings find it progressively harder to find the copyright owner to secure permission — particularly when copying is necessary because, for example, a book is out of print. The owner may have moved, gone out of business, sold his copyright, or died.¹⁷⁴ Yet to facilitate the copying of old writings is particularly important, for an old writing that someone wishes to copy is likely to have unique merit or to be needed for research or education — areas likely to contain spill-over benefits.

Third, to extend copyright protection may restrict competition between copyrighted works and works that would otherwise fall into the public domain. Owners of music copyrights, for example, do not compete vigorously in the sale to radio and television stations of licenses to perform their works. Rather they have combined in organizations, such as ASCAP,¹⁷⁵ that normally sell blanket licenses providing a buyer with the right to play all ASCAP works for a single price.¹⁷⁶ ASCAP's prices are controlled by a federal court (as a result of an antitrust action), but its revenues are also limited by the ability of its customers to substitute music in the public domain for the music it licenses.¹⁷⁷ To remove music from the public domain by extending copyright protection will make this substitution more difficult;¹⁷⁸ it should thereby allow ASCAP to increase its revenues at the expense of its licensees and — insofar as they can pass on this added cost — at the expense of the public. Yet, as noted earlier,¹⁷⁹ copyright law is intended only to protect an author from

¹⁷⁴ Cf. *Chamberlain v. Feldman*, 300 N.Y. 135, 89 N.E.2d 863 (1949).

¹⁷⁵ For an explanation of the workings of ASCAP (the American Society of Composers, Authors, and Publishers), as well as two other, considerably smaller, licensing organizations, BMI (Broadcast Music, Inc.), and SESAC (Society of European State Authors and Composers), see Finkelstein, *Public Performance Rights in Music and Performance Right Societies*, 7 COPYRIGHT PROBLEMS ANALYZED 12-21 (1952) [hereinafter cited as Finkelstein].

¹⁷⁶ *Id.* at 14. The sale of the blanket license does not eliminate the possibility of competition between works, for buyers may also buy licenses for individual songs.

¹⁷⁷ Stations may obtain a "program license" which allows them to play ASCAP music only on "those programs in which music in the Society's repertory is performed." *Id.* at 14.

¹⁷⁸ It will not make substitution *impossible*, for the possibility of application for an individual performance license exists. See note 176 *supra*. Yet the complexity and expense involved in numerous applications for individual licenses will make substitution more difficult if a work is removed from the public domain.

¹⁷⁹ See p. 318 *supra*.

competing sales of his own work; it was not designed to limit competition from the works of others.¹⁸⁰ As copyright is extended in time, then, it seems probable that the harm that it causes will grow more rapidly than will the benefits that it produces.¹⁸¹

Attempts to justify extension on grounds other than as an incentive to creation or production seem equally weak. Arguably, for example, extension will help the author or his heirs prevent the garbling of a work. But there is no indication that garbling of "classics" is now a serious problem. And this rather unimportant consideration is overshadowed by the marginally more substantial fear that extension would allow the "author-hating" child to prevent further publication of his parent's works. Boswell's son believed his father's relation with Dr. Johnson demeaning and might have suppressed the famous journal. Indeed, an heir has recently used the power that copyright provides to prevent a distinguished biographer from publishing a work about Kipling.¹⁸²

It is possible that to substitute a period of "life plus fifty years" from a work's *creation* for the present fifty-six years from a work's *publication* will avoid the knotty legal problem of determining when publication takes place.¹⁸³ Yet, as Professor Kaplan

¹⁸⁰ If buyers are at least equally satisfied with *A*, a work in the public domain, calling forth the production of *B* by restricting *A* wastes resources, at least when such a restriction is greater than necessary to induce *A*'s initial production.

¹⁸¹ This conclusion may be questioned on the ground that the harms that extension may cause must be discounted because they occur in the future, while the benefits it may produce, in terms of additional books induced, will occur in the present. The theoretical problems involved in comparing harm and benefit across generations are considerable. See, e.g., MUSGRAVE, *supra* note 90, at 562-65; Eckstein, *A Survey of the Theory of Public Expenditure Criteria*, in NATIONAL BUREAU OF ECONOMIC RESEARCH, *PUBLIC FINANCES: NEEDS, SOURCES OF UTILIZATION* 453 (1961). Discounting, even if appropriate, should not, however, affect this conclusion significantly. For one thing, the competitive harms of extension will occur now for books about to fall into the public domain. For another, extra production, if it occurs, would take the form of books that, for the most part, last only a few years. Extra harm, though it occurs in the future, concerns classics (and other particularly useful books), and it therefore will reoccur in each additional year of extension. If the term is extended from 56 to 76 years, for example, the harms discussed may be repeated in each of the additional twenty years. This factor will tend to offset the discount. Assuming a social discount rate of three percent, three years worth of an additional benefit of \$1 per year is now worth \$2.83; twenty years of \$1 worth of additional harm per year, beginning 56 years in the future, is now worth \$2.84. See Eckstein, *supra*.

¹⁸² Letter from F. Birkenhead to Stephen Breyer, Nov. 12, 1970.

¹⁸³ If this is the reason for extension, the longer period would be necessary because statutory protection would have to apply not only to published works, but also to those that the author prefers never see the light of day.

has pointed out, the definitional problem is not overwhelming.¹⁸⁴ In fact, in its 1961 report on the revision bill, the Copyright Office recommended that the term be based upon the time of dissemination rather than the author's life because, for those wishing to reproduce a book, "the death date of authors who are not well known would often be difficult to ascertain."¹⁸⁵

The revisors also may have been moved by the plight of a few aged authors or composers attempting to live on royalties from works written in their youth. The aged writer himself presents a more sympathetic case than does his middle-aged orphan. But one need only recognize that such a writer would, for example, have to be over eighty and to have written his last popular work before the age of twenty-four to see that the poor, but doddering, prodigy is an unusual case.¹⁸⁶ Nor is it obvious why it should disturb us more to see a writer outlive his fifty-six year copyright than an inventor, his seventeen year patent. If simple income support is desirable, other forms of economic assistance may be more appropriate.

A final argument — that extension will bring America into conformity with many foreign countries — is often made, but I have found no clear explanation presented of the advantages of uniformity per se.

Some other nations with longer periods of protection have tried to minimize the harm involved through a series of ad hoc devices. England, for example, recognizing the possibility that competition *between* books may prove less effective at holding down the price of "classics," provided in 1911 that during the last twenty-five years of the copyright period anyone can reproduce a copyrighted work upon payment of a ten percent royalty to the copyright owner.¹⁸⁷ Unfortunately Parliament recently repealed this provision on the ground that it was not used, overlooking the fact that its mere presence — the threat of its use — may have kept book prices lower than they would have been otherwise.¹⁸⁸ Certain other countries mitigate the difficulty of contacting owners of old copyrights by providing that a man may reproduce a copyrighted work after he has made a reasonable

¹⁸⁴ KAPLAN 96.

¹⁸⁵ Register of Copyrights, *supra* note 166, at 48.

¹⁸⁶ This is not to say that no such cases exist. The House Committee, for example, heard from Mr. Will Dillon, 87 years old, who received royalties from the song, "I Want a Girl Just Like the Girl that Married Dear Old Dad," written when Mr. Dillon was 33. *Hearings on the Copyright Law Revision Before Subcomm. No. 3 of the House Comm. on the Judiciary*, 89th Cong., 1st Sess., ser. 8, pt. 1, at 228-29 (1965).

¹⁸⁷ Copyright Act of 1911, 1 & 2 Geo. V, ch. 46 § 3.

¹⁸⁸ Copyright Act of 1956, 4 & 5 Eliz. 2, ch. 74, sched. 7.

effort to contact the copyright owner and failed.¹⁸⁹ Another country requires payment to extend copyright beyond an initially brief term.¹⁹⁰ Such a requirement might provide additional protection for commercially valuable works while allowing those with little value to fall into the public domain, permitting their use for research without impediment. Such measures, while perhaps unnecessary if the term of copyright is relatively short, are surely worth considering to offset the harms of any extension.

If, as suggested in Part II, the basic case for copyright protection of books is weak, and the harms that it causes grow more rapidly than the benefits that it yields, should not the Copyright Law Revision propose measures to cut back the term of copyright, rather than to extend it? There is little to suggest any net gain in extending book protection beyond the time horizon that the publisher uses when he decides whether a publication can pay its way, and this time horizon probably does not exceed fifteen to twenty years even for the most complex encyclopedia.¹⁹¹ In any event, extension beyond fifty-six years seems clearly unjustified, particularly without any of the mitigating devices mentioned above.¹⁹²

B. Photocopying

Copyright owners have recently sought both legislative¹⁹³ and

¹⁸⁹ See, e.g., Kase, *Copyright in Czechoslovakia: The New Copyright Statute of 1965*, 14 BULL. OF THE COPYRIGHT SOC'Y 28, 45 (1966).

¹⁹⁰ See Copyright Law of Honduras, 1 UNESCO, COPYRIGHT LAWS & TREATIES OF THE WORLD (1956-1970) (1970).

¹⁹¹ See p. 325 *supra*.

¹⁹² The Revision Bill goes even further than I have suggested in text and makes extension retroactive; moreover, pending passage of the Bill, Congress has extended the life of all copyrights that would otherwise expire. It seems totally unjustified, however, to allow holders of *present* copyrights to benefit by the extension. In the case of presently copyrighted works, additional incentive to produce is irrelevant. Moreover, unlike Mr. Dillon, *see* note 186 *supra*, many creators may not own the copyrights in their own works. Publishers or others may have obtained the copyright directly by purchase or by assignment; or, like the publisher of *Little Black Sambo*, they may have bought the copyright at a bankruptcy sale. To apply extension retroactively is simply to provide a diverse collection of heirs, estates, publishers, and possibly a few aged authors with a windfall gain at the expense of readers.

¹⁹³ Publishers, for example, have sought the inclusion of language in the Revision Bill that would make it clear that photocopying requires permission when it may interfere with revenue that the publisher would otherwise receive. *See, e.g.*, Statement of Williams & Wilkins Co., *Copyright Law Revision, Hearings Before Subcomm. No. 3 of the House Comm. on the Judiciary*, 89th Cong., 1st Sess., ser. 8, pt. 3, 1511-18 (1965); H.R. REP. NO. 2237, 89th Cong., 2d Sess. 58-66 (1966). *See generally* G. GIPE, *NEARER TO THE DUST* (1967).

judicial¹⁹⁴ action that would provide more effective protection against unauthorized photocopying. At present, copyright law is unclear on the subject of the xerox machine. The elusive, judicially created doctrine of "fair use," according to the Register, would allow "a reasonable portion of a copyrighted work . . . [to be] reproduced without permission when necessary for a legitimate purpose which is not competitive with the copyright owner's market for his work."¹⁹⁵ But what is a "reasonable portion?" And when is a purpose "not competitive with the copyright owner's market?" Most librarians have interpreted this doctrine liberally and have acted (without hindrance) as if it allows a user to make a single copy of the *whole* of a magazine article or book chapter — at least for research purposes.¹⁹⁶ Many publishers, however, hold the more restrictive view that ordinarily only *parts* of magazine articles can be copied without permission,¹⁹⁷ at least if the copying is done by machine rather than by hand.¹⁹⁸ The Revision Bill's language is ambiguous on the question,¹⁹⁹ but the House Committee's report on the Bill seems to adopt the publishers' view of the law.²⁰⁰

Although photocopying is widespread, it may still prove pos-

¹⁹⁴ The Williams & Wilkins Co. has brought suit to determine that the present photocopying practices of the National Library of Medicine infringe their copyrights. See Association of Research Libraries, Newsletter No. 28, March 12, 1968, at 1. This suit appears to be the first that will directly place at issue the legality of making photocopies of whole articles available to users or to subscribing libraries.

¹⁹⁵ Register of Copyrights, *supra* note 166, at 24.

¹⁹⁶ The CIRC concludes in its recent survey of 66 libraries that most librarians ". . . believe the making of a single copy of an article from a periodical or . . . a book . . . is within the area of 'fair use.'" CIRC ii, 80.

¹⁹⁷ This characterization attempts to summarize many different "restrictive" interpretations of "fair use." The restrictive interpretations flow in part from a general premise that any copying that restricts the publisher's revenue is not a "fair use." See generally NIMMER § 145 (1967); cf. H.R. REP. NO. 2237, 89th Cong., 2d Sess. 64 (1966).

¹⁹⁸ It has long been argued that copying by hand for private study or research purposes is a "fair use." See Cohen, *Fair Use in the Law of Copyright*, 6 ASCAP COPYRIGHT LAW SYMPOSIUM 43, 58 (1965). And, under the "Gentlemen's Agreement" of 1939, machine copying is legal when it substitutes for hand copying. See 2 J. OF DOCUMENTARY REPRODUCTION 29 (1939). Yet much present machine copying does not substitute for hand copying, for in the absence of the xerox machine, many users would not take the time to copy by hand.

¹⁹⁹ The Revision Bill enacts language stating that factors relevant to fair use are: (1) the purpose and character of the use, (2) the nature of the copyrighted work, (3) the amount and substantiality of the portion used in relation to the copyrighted work as a whole, and (4) the effect of the use upon the potential market for or value of the copyrighted work. S. 543, 91st Cong., 1st Sess. § 107 (1969). As the Scottish preacher told his congregation when delivering a sermon on an abstruse theological question, "Now, bretheren, wee cum to a verra great deeficulty: wee shall look it squarely in the face and pass on."

²⁰⁰ The Report states that "a use which supplants any part of the normal

sible to enforce the publishers' views — at least if the law is clearly on their side. The CICIP survey states that "total annual copying of published material . . . is very largely, but not entirely, library copying."²⁰¹ There are roughly 15,500 public school systems with centralized libraries,²⁰² and 22,500 other libraries in the United States.²⁰³ It may be possible to sell reproduction permissions to these libraries through a publishing clearinghouse modeled after ASCAP, a music clearinghouse, which now licenses more than 30,000 radio and television stations, nightclubs, and others to perform copyrighted music for profit.²⁰⁴ Such a clearinghouse could grant schools and libraries a "blanket permission" to reproduce all the works of its author and publisher members in return for a payment, which it would then distribute among those members. Of course, nearly eighty-five percent of ASCAP's revenues come from 4500 to 6000 radio and television stations,²⁰⁵ but, similarly, most reproduction of copyrighted works is probably done in the 2900 college libraries²⁰⁶ or the 900 public libraries that serve populations of more than 35,000.²⁰⁷

To show that enforcement of the publishers' views may be

market for a copyrighted work would ordinarily be considered an infringement Where the unauthorized copying displaces what realistically might have been a sale, no matter how minor the amount of money involved, the interests of the copyright owner need protection." H.R. REP. NO. 2237, 89th Cong., 2d Sess. 64 (1966).

²⁰¹ CICIP 70. The Williams & Wilkins Co. notes that photocopying has not interfered with subscriptions to clinical journals to the same extent that it has interfered with subscriptions to academic journals. A spokesman for the company explained this fact on the ground that clinicians are more likely "to afford the convenience of subscriptions, while academicians usually have greater access to copying machines." *Project, New Technology and the Law of Copyright: Reprography and Computers*, 15 U.C.L.A. L. REV. 931, 945 (1968) [hereinafter cited as *UCLA Project*].

²⁰² AMERICAN BOOK PUBLISHERS COUNCIL, THE BUCK HILL FALLS REPORT 19 (1966).

²⁰³ U.S. BUREAU OF THE CENSUS, STATISTICAL ABSTRACT OF THE UNITED STATES: 1969, at 123.

²⁰⁴ *Special Report: ASCAP: Oldest, Biggest, Toughest*, BROADCASTING 148 (Mar. 15, 1965).

²⁰⁵ *Id.* A letter from Herman Finkelstein, ASCAP's General Counsel, states that approximately 85% of ASCAP's revenue is obtained from 15% of its licensees. BROADCASTING gives the latter figure as 20%. Presumably ASCAP's primary sources of revenue are commercial AM radio stations and commercial TV stations. In 1968 there were 4,289 commercial AM stations and 835 commercial TV stations in the United States. U.S. BUREAU OF THE CENSUS, STATISTICAL ABSTRACT OF THE UNITED STATES: 1969, at 499.

²⁰⁶ This figure includes junior college, college, and university libraries. U.S. BUREAU OF THE CENSUS, STATISTICAL ABSTRACT OF THE UNITED STATES: 1969, at 123.

²⁰⁷ AMERICAN BOOK PUBLISHERS COUNCIL, THE BUCK HILL FALLS REPORT 21 (1966).

feasible, however, does not show that it is desirable. Indeed, it is difficult to believe that the need for enforcement is sufficiently great to warrant the considerable costs, harms, and complexities that would be involved.

A clearinghouse, while far less costly than a system of individual negotiation,²⁰⁸ would still involve heavy administrative costs. In 1967 ASCAP spent roughly \$9 million to collect gross domestic revenues of \$49 million.²⁰⁹ A publishing clearinghouse — as does ASCAP — would have to sell licenses, monitor unlicensed copiers to detect violations, bring infringement actions where piracy is found, make random samples to determine the relative frequency with which works are copied, and distribute collected income to numerous writers and publishers.²¹⁰ It is difficult to believe that a publishing clearinghouse would cost any less to operate than ASCAP; and, unless it is prepared to ignore infringements by smaller libraries and school systems, it might cost considerably more. At the same time, it seems unlikely that a publishing clearinghouse could collect as much revenue as ASCAP. Although the number of licensees may be roughly comparable, it is doubtful that schools and libraries would be willing, or could afford, to pay as much for the right to make photocopies as commercial radio and television stations pay for the right to play copyrighted music.²¹¹

²⁰⁸ See p. 317 *supra*. It is also possible that transactions costs will be reduced by the publishers adopting the simple device of charging more for "special" copies carrying with them the right to make a certain number of photocopies. The only administrative expense involved would be that of affixing in each "special" copy a notice stating that the buyer also receives the right to make a certain number of photocopies. Unfortunately, one cannot rely upon publishers coming to this solution on their own, for some publishers may make more money by refusing to publish "special" editions. Without such an edition, some potential copiers may negotiate with the publisher individually or buy regular copies instead. In either case, the publishers will be able to discriminate roughly in price, receiving more money from those to whom the right to copy is more valuable. Special copies would allow publishers to discriminate in price *between* the class of persons wishing to reproduce and the class of persons not wishing to reproduce; but *without* special copies, the publisher may be able to discriminate in price *among* members of the former class. One cannot say in advance to which publisher the latter advantage will prove more important than the former.

²⁰⁹ *ASCAP: A Record Year but Too Much 'Talk'*, BROADCASTING 51 (March 4, 1968). ASCAP is by far the largest of the three composers' organizations.

²¹⁰ See generally Finkelstein, *Public Performance Rights in Music and Performance Right Societies*, 7 COPYRIGHT PROBLEMS ANALYZED 1 (1961); Finkelstein, *ASCAP as an Example of the Clearing House System in Operation*, 14 BULL. OF THE COPYRIGHT SOC'Y 2 (1966).

²¹¹ If we take as accurate the figures revealed by the CACP's survey in depth of six libraries, roughly 150 million copies of copyrighted articles and books were made in 1969. See CACP at 62, 71. A charge of 33 cents per copy would be necessary to raise \$50 million, and this amount is unlikely to provide significant

The fact that revenues may not be large does not mean that permissions' prices will prove harmless. It is entirely possible that a xeroxing charge high enough to yield a moderate revenue will at the same time discourage a sizable amount of photocopying, depriving users of the considerable benefits that the xerox machine can bring.²¹² One cannot rely upon competition between publishers to hold down the price of permissions, for "blanket licenses" and joint price determinations make competition between publishers in the sale of reproduction rights most unlikely.²¹³ In fact, if publishers can agree upon the price they will charge for this right, they may be able to extract much of the direct economic benefit that educational xeroxing promises its users.

This anticompetitive problem is now worthy of note, it may become far more serious in the future,²¹⁴ and there seems no very satisfactory way of mitigating its effects. Experience with ASCAP, which since 1941 has been regulated under an antitrust

additional revenue for individual authors if one accepts the CACP's conclusion that most copying done is of *different* articles. Of the 3,816 titles copied in the six library sample, 2,946 were different. CACP 60. At the same time, a charge of 33 cents or more per article copied may be high enough to discourage considerable photocopying and thus reduce the publishers' revenue below the \$50 million figure.

²¹² These benefits include the aid to research and education that widespread dissemination of information produces. Many of these benefits are "spillover" benefits. Thus the fact that libraries and schools may not pay much for the right to xerox does not show that their xeroxing is not of great value for society.

²¹³ The publishers would maximize their revenue by setting a price that is higher than would be set if they competed in selling the right to make photocopies. In a joint association they may set this price. Even if prices were regulated, as with ASCAP, to avoid antitrust problems, the competitive price can only be approximated.

²¹⁴ We may in the next decade see the development of automatic information systems. These systems would store printed information on microfilms or microfiches (which are film transparencies of, for example, playing card size each of which may store one hundred or more printed pages). Subscribers located anywhere in the country might use a computer, via terminals located in home or office, to discover in which library relevant information is kept. The libraries storing information would be interconnected by means of communications wires, which would also run to the homes or offices of subscribing viewers. The subscriber might use a computer terminal in his home or office to seek relevant information; the computer would automatically contact the appropriate library, and the librarian would insert the appropriate microfiche in a device that would cause its contents to appear on the subscriber's home or office viewing screen. Since microfiche is an essential ingredient in this system, a clearinghouse that allows all or most publishers to combine in setting a price for the right to make a microfiche copy may allow those publishers to extract much of the economic benefit that the development of such a system promises. For more detailed and interesting discussions of future information dissemination systems, see 1967 *Senate Hearings* 547-81; *CATV and Microforms*, Speech by H. Hilton, 19th Annual Convention of the National Microform Association, April 28, 1970.

decree,²¹⁵ suggests that to require a clearinghouse to offer individual, as well as blanket, licenses does little to promote competition, for buyers rarely seek individual licenses.²¹⁶ To ask an administrative agency or a court to regulate or to supervise clearinghouse prices invites complex and nearly irresolvable disputes over what price is "fair."²¹⁷ And to have Congress provide by statute for a compulsory license at a fixed fee may embed what soon becomes an arbitrary sum into the law for many years.²¹⁸

In sum, to adopt the publishers' views on "fair use" would impose high transaction costs and the threat of prices that would discourage some photocopying. At the very least it would require school and library users, who are typically in need of money themselves, to spend a large amount of money to administer a system the sole function of which is to transfer an additional large amount of money from the same schools and libraries (or their users) to authors and publishers. Surely before adopting these views a substantial showing should be made that these expenditures are in fact needed to maintain writing production.

The analysis of book production previously made, however, must raise doubts that additional copyright protection is needed

²¹⁵ The anticompetitive problems that a clearinghouse creates are analogous to those created by patent pooling, *see, e.g.*, *Hartford-Empire Co. v. United States*, 323 U.S. 386 (1945); *Standard Oil Co. of Indiana v. United States*, 283 U.S. 163 (1931), and the package licensing of patents, *see Automatic Radio Mfg. Co. v. Hazeltine Research, Inc.*, 339 U.S. 827 (1950). The courts in these cases have used a "rule of reason" to assess antitrust legality. Presumably ASCAP felt that there was enough risk that the evils potentially present in its price-fixing power would outweigh the potential benefits to enter into a consent decree under which the court may supervise ASCAP's prices. *See United States v. ASCAP*, 341 F.2d 1003 (2d Cir. 1965); [1940-43] CCH TRADE CAS. ¶ 56,104; [1950-51] CCH TRADE CAS. ¶ 62,595; [1960] CCH TRADE CAS. ¶ 69,612.

²¹⁶ The antitrust decree prevents ASCAP from prohibiting a copyright owner/member from selling an individual license. [1950-51] CCH TRADE CAS. ¶ 62,595, at 63,752. Yet as a practical matter the negotiation of individual licenses is difficult and relatively expensive (the same transaction costs as for a blanket license may be involved). Moreover, the individual copyright owner may recognize that competition in the sale of individual licenses can lower total returns by undermining the jointly set blanket license price. *See Finkelstein, Public Performance Rights in Music and Performance Right Societies*, 7 COPYRIGHT PROBLEMS ANALYZED 1, 13-15 (1961).

²¹⁷ The difficulties of determining a "fair" price in this area were recognized in 1838 by A.C. Renouard. *See A. RENOUARD, TRAITE DES DROIT d'AUTEURS*, vol. 1, § IX, at 463 (1838). One is, in essence, asking a court to guess what price would be reached by bargaining were the market relatively free.

²¹⁸ In 1909 Congress provided a compulsory license for the reproduction of musical works and set a two cent per copy maximum fee. This provision has remained in the statute unchanged for more than sixty years. For other variations, *see U.C.L.A. Project 950-73; Committee to Investigate Copyright Problems, First Annual Report*, 10 BULL. OF THE COPYRIGHT SOC'Y 1 (1960).

to avoid the loss of frequently copied works — which, for the most part, are scholarly journals.²¹⁹ In many cases photocopying will not seriously interfere with journal revenues, either because it stimulates greater interest in the journal²²⁰ or because publishers can make up for a decline in the number of subscriptions by raising their prices.²²¹ After all, if xeroxing allows libraries to buy fewer subscriptions, it also makes those few relatively more valuable. Moreover, many groups of buyers will be able

²¹⁹ The CICIP study concludes that "library copying is mainly from journals and mainly from nonprofit journals." CICIP 68. The ratio of journal titles to book titles copied was more than nine to one. *Id.* at Table I. Scientific and technical journals were the main materials copied. *Id.* at 67, 77. The UCLA Project suggests that the figures upon which the CICIP bases these conclusions may have been biased to some extent, though, by the nonrandom nature of the sample. UCLA Project 1026-27.

In addition to scholarly journals, one suspects that most copying done is of textbooks and technical monographs or advanced treatises. I doubt, for reasons mentioned in Part II, *see* pp. 319, 321 *supra*, that texts need give us great concern. Advanced treatises and monographs raise more difficult problems. Curtis Benjamin, the Chairman of the Board of Directors of McGraw-Hill, has written that the photocopying phenomenon, combined with the increasing specialization of the sciences, has led to a fall in the average number of copies of such books sold. In 1943-47 the number of copies sold of specialized scientific books published by McGraw-Hill averaged 4,843; in 1963-67, it averaged 3,761. Benjamin, *What Can Be Done to Save the Monograph?*, SCIENTIFIC RESEARCH 32 (Sept. 16, 1968). Mr. Benjamin, interestingly enough, when faced with a genuine threat to production, suggests not increased copyright protection but rather:

"sale to libraries, research organizations, and other institutions under an advance subscription scheme whereby the purchasers would have contracted rights [of reproduction] . . . This scheme would mean one-shot publishing at much higher prices, of course. The publisher would think of producing and selling, for example, no more than 500 copies at \$100 per copy (with full reproduction privilege), rather than 5,000 copies at \$10 per copy."

Id. at 33.

Mr. Benjamin's solution would seem appropriate whether or not the copyright law allows relatively free photocopying privileges. He adds that he has "proposed this scheme to a number of scientists and research librarians. All but a few backed away sharply." But suppose the alternative is no publication? The libraries should not find buying a book under Mr. Benjamin's scheme significantly more difficult than at present. *Id.* at 33.

²²⁰ Authors will, of course, approve the wider circulation that xeroxing gives their work. Yet this benefit will be offset to some extent by the fact that insofar as single copies of selected articles, rather than whole magazines, are circulated, "browsing" through other unrequested articles in a magazine may decline.

²²¹ In fact, since most of these journals may receive much of their revenue from other than subscriptions, a decline in subscriptions will require a less than proportional increase in subscription prices. A National Science Foundation survey in 1962 of 262 representative scientific journals revealed that 211 were published by nonprofit scientific societies, 18 were published by university presses, and 33 were published by commercial publishers. The "society" journals received 41% of their income from subscription sales; and the "university press" journals, 56%. (No figure is given for the "commercial" journals.) NATIONAL SCIENCE FOUNDATION, CHARACTERISTICS OF SCIENTIFIC JOURNALS 1949-1959 at 2, 11, 12 (1964).

to find other ways of channeling more money to journals whose existence is financially threatened. Many such journals are published or supported by learned societies, universities, or foundations, which either consist of, or represent, the very persons who subscribe to these journals and who benefit from being allowed to copy freely.²²² Such communities of subscribers should not find it difficult to increase society dues or the university's subsidy, if necessary. Doing so should ordinarily not hurt them financially, for, considered as a community, they would pay no more *totally* for the journal than they do at present, and they would receive from it at least as much benefit. And groups of academics and professionals should be sufficiently sensitive to the need for journals in their fields to minimize "freeloading" difficulties. Finally, an increase in the already enormous subsidy that the government provides for disseminating information is at least a theoretical possibility,²²³ though whether the photocopying problem by itself requires an increase in the subsidy is doubtful.²²⁴

Of course, the considerations just mentioned are not sufficient to guarantee that *all* journals will continue in business. Yet, the loss of a few would seem preferable to spending \$10 million or more on an administrative scheme designed to impose a charge on

²²² *Id.* at 11.

²²³ See p. 307 & note 100 *supra*.

²²⁴ Technological developments may lead to replacement of the journal as an important medium for disseminating scientific and technical information. The government may expand the dissemination of many of the results of the research work that it finances by publishing "single copies" of individual articles separately. See UCLA Project 946. Automatic information systems and clearinghouses may also replace the journal as an important dissemination medium. See REPORT OF THE TASK GROUP ON DISSEMINATION OF INFORMATION, RECOMMENDATIONS FOR IMPROVING THE DISSEMINATION OF FEDERAL SCIENTIFIC AND TECHNICAL INFORMATION (Dec. 1969).

If a federal subsidy is felt desirable, it would seem wiser to raise the funds from general tax revenues than from a tax on copying or, as has sometimes been proposed, from a tax on xerox machines. See UCLA Project 973. Since much photocopying produces considerable spillover benefits in the form of additional research or improved education, it is not unfair to impose the tax upon the general taxpayer rather than upon the immediate copier. Moreover, a tax on xerox machines will be hard to administer fairly, as some machines are used more extensively to copy copyrighted materials than others. And some photocopying is done in other ways. See CACP 84. Attempts to correlate more finely the type of copying done with payment to the copyright owner (whether or not through the government) will begin to introduce the high transaction costs discussed above.

A system that relies upon government subsidy, regardless of how the revenue is raised, runs some risk of censorship when the money is distributed. Yet at least for journals in the hard sciences the risks are probably not much greater when dissemination is at issue than when the question is one of basic research, much of which the government now pays for. See note 100 *supra*.

photocopying that may inhibit the dissemination of copyrighted works. It would seem sensible, then, to embody the librarians' views, rather than the publishers' views, in the copyright law. Congress might simply provide that the copying of periodical publications and extracts from books by an individual for his own use does not violate the copyright law provided that he does not make more than one copy within a certain period of time. Such an exemption would eliminate the law's existing ambiguity about "fair use" — a confusion which invites unfairly selective attempts at enforcement and which may breed disrespect for the copyright law in other areas.²²⁵

To guard against the risk that advancing technology and changing economic circumstances will make this exemption obsolete, Congress might provide a means for adjusting periodically the boundaries of the "fair use" exemption. The Revision Bill proposes the creation of a commission to investigate copyright law and new technology.²²⁶ Such a commission might more extensively examine particular sorts of copying and particular types of copyrighted material. It could gather facts allowing it, for example, to estimate transactions costs, to determine the state of competition between works, and to assess possible alternative methods for securing publishing revenue. The commission might thereby judge with some accuracy the costs of increasing or rolling back copyright protection. It should recommend changes in the scope of "fair use" to Congress, and were it to become permanently embedded in copyright law, its recommendations might take effect subject to congressional veto.²²⁷

²²⁵ If necessary, the exemption could be modified to protect textbook publishers. The British copyright law, which contains a similar exemption, grants such protection by restricting the exemption to copying done "for research or private study." See Copyright Act of 1956, 4 & 5 Eliz. 2, c. 74, § 6(1); *THE SOC'Y OF AUTHORS & THE PUBLISHERS ASS'N, PHOTOCOPYING AND THE LAW* (1965). Yet in light of the weakness of the basic case for copyright in texts, one wonders if this limitation is necessary. The British law also restricts the exemption to copying done in libraries or educational institutions, but this further restriction is of dubious value. To enforce the copyright law against the vast array of private copiers seems so impractical that to allow it invites the unfairness of highly selective enforcement. And, as previously mentioned, most copying of published materials is library copying.

²²⁶ Title II of the Copyright Revision Bill provides for the creation of a National Commission on New Technological Uses of Copyrighted Works. Its purpose is to

study and compile data on . . . the reproduction and use of copyrighted works of authorship . . . both in conjunction with automatic systems capable of storing, processing, retrieving, and transferring information, [and] . . . by various forms of machine reproduction . . . [and to study] the creation of new works by the application or intervention of such automatic systems or machine reproduction.

S. 543, 91st Cong., 1st Sess., tit. II (1969).

²²⁷ Such a procedure is now used for changes in the Federal Rules of Civil

C. Research with Computers

The discussion of photocopying shows how we can shape an exemption to minimize the burden of copyright protection once we have identified the more serious costs that accompany copyright. A similar, but more simple, example of the ease with which an appropriate exemption can be tailored to deal with high transactions costs arises from the use by many scientists and scholars of copyrighted material with computers for research purposes.

From the point of view of copyright law, research with a computer can differ from research without it in an important respect, namely, the computer operator may have to make a "computer-readable" copy of the material for insertion into the machine, and the machine may make another copy of the material when it prints out responses. The scientist without a computer may simply refer to a copyrighted book of chemical equations as he does his work; yet the scientist who does the same work with a computer will have to insert the equations into the computer first, and the computer may also print some of them when it delivers its answers. Unless this copying is held to constitute "fair use," the Copyright Revision Bill would require individual scientists and scholars to contact copyright owners for permission when they use copyrighted material along with a computer.

Scientists see the "permission" requirement as imposing a time-consuming, cumbersome, and expensive burden that would threaten research by imposing delay. Professor Anthony Oettinger, Director of the Harvard University Computing Center, testified that, under the provisions of the Revision Bill: ²²⁸

I would have not only to acquire and evaluate materials but, in each instance, *before* experimenting with them, seek out the owner of a copyright, if any, make formal requests for permission to use the material, pay royalties if any are due, etc. All this before any material could actually be used and, in fact, before I could find out whether or not the material was useful! The delays, the frustrations and the chaos inherent in such a process now seem so formidable that if the bill were passed in its present form I would be tempted to return to the safer occupation of copying out manuscripts with a goose quill pen.

On the other hand, publishers fear that to relax copyright requirements in this area would allow computers to be used as

Procedure. Changes are proposed by the Supreme Court and take effect within a given period of time unless Congress acts.

²²⁸ Statement of Professor Anthony G. Oettinger, 1967 *Senate Hearings*, pt. 2, at 581, 588.

libraries in the sense that one copy, fed into a computer, will serve many users who might otherwise have bought their own copies.²²⁹ A "permissions" requirement, by allowing publishers to extract a fee for the computer use, would help to safeguard their revenues.

It is not difficult, however, to write an exemption that would help the scientist, while minimizing harm to the publisher. Such an exemption would allow individuals and small groups to store copyrighted material in computers and to retrieve it without the owner's permission provided that the storage and retrieval form part of a project of research or private study in which the individual or group is engaged. The "small group" limitation prevents the computer from serving as a library. At the same time the smaller the group the more likely that the administrative costs of securing permission are high in relation to the need for the copyrighted material. Thus the exemption focuses on that area where copyright's harms are greatest and benefits smallest.²³⁰

²²⁹ Because of the cost of storing information in electronic form within a computer (or even on tapes or discs that can be used easily in conjunction with the computer), it seems unlikely that computers will actually replace libraries to any significant degree within the next decade or two. See generally *UCLA Project* 993-1000; Greenbaum, *Copyright and the Computer*, 114 CONG. REC. 17042 (1968). Nonetheless, the storage of equations or similar information now contained in copyrighted manuals may replace some sales of these manuals if large numbers of potential buyers have access to the computer.

²³⁰ The question of when computer operators should be entirely free of copyright strictures is an important one deserving of more attention than it has received. Much of the "computer/copyright" discussion so far has focused on what I believe is a far less significant problem, namely whether copyright liability should attach when material is put *into* a computer or when it *emerges* — whether the copyright toll should attach at the "input" or the "output" stage. See, e.g., *UCLA Project* 1004-17. I doubt that the "input/output" problem is very important because its resolution is unlikely to affect *when* bargaining between user and owner takes place (even if permission is required only for output, the computer operator will, as a practical matter, have to seek it before the material is put into the computer). Neither should it affect the *amount* of payment (which depends upon the parties' bargaining power) nor the ease of detecting violations (which should depend upon tracking output in either case).

The "input/output" problem rose to prominence only when EDUCOM, an organization of university representatives interested in automatic information systems, discovered that to attach a copyright toll at the "input" stage threatened to destroy portions of the exemptions that the copyright law provides for various nonprofit institutions. The Bill, for example, exempts the "display of a work by instructors . . . in the course of face-to-face teaching." S.543, 91st Cong., 1st Sess., § 110(1) (1969). This language may exempt certain displays of, say, copyrighted tables when they emerge from a computer for classroom use; yet the exemption will prove meaningless if a copyright toll attaches when the tables are put into the computer. See *The Copyright Revision Bill in Relation to Computers*, 1967 *Senate Hearings*, pt. 2, at 570, 576. On the other hand, publishers argue that to attach a toll only at the "output" stage would free from liability material put

Moreover, in setting what must be an essentially arbitrary upper limit to the size of a "small group," Congress should not fear to be generous to the scientist. Most of the copyrighted material that will be stored in computers in the near future will consist of books of tables, manuals of equations, compilations of abstracts, and other highly scientific and technical works.²³¹ For reasons pointed out at p. 336 above, in the case of such works a decline in the average number of copies that each work sells is not likely to interfere significantly with their production.²³²

D. Computer Programs

The Revision Bill would provide copyright protection for the first time for computer programs.²³³ Those familiar with com-

into a computer that *never* emerges in a form that would infringe a copyright. (For example, copyrighted mathematical tables might emerge only in the form of responses to questions.)

In fact, it is possible to modify the law to deal with either of these problems. A toll placed upon "input" would require tailoring congruent exemptions so that they apply at both the "input" and the "output" stage. A toll on "output" would require carefully redefining "infringement" so that computer operators cannot use copyrighted works without paying a toll where such payment is economically desirable. Neither of these adaptations would work perfectly: a toll that attaches to input would still raise problems when the input is to be put to some uses that are exempt and others that are not; a toll only upon output may require complex and difficult work in redefining "infringement." Since I believe the former problem more serious, I would prefer to see the toll attach only at the output stage. But, it is difficult to believe that an adapted version of either solution will work serious harm on those who oppose it. The Revision Bill's present solution, however, appears to be a toll on "input" *without* adaptation—a solution that may harm schools. See S. 543, 91st Cong., 1st Sess. § 101 (1969).

²³¹ See note 229 *supra*.

²³² The publisher might raise the price per copy, for example, or use other methods such as those suggested to safeguard revenues.

²³³ The Revision Bill's definition of what is copyrightable would seem broad enough to include computer programs. Section 102(a) states:

Copyright protection subsists . . . in original works of authorship fixed in any tangible medium of expression, now known or later developed, from which they can be perceived, reproduced, or otherwise communicated, either directly or with the aid of a machine or device.

Section 101 defines "literary works" to include "works other than audiovisual works, expressed in words, numbers, or other verbal or numerical symbols or indicia, regardless of the nature of the material objects . . . in which they are embodied." S. 543, 91st Cong., 1st Sess. § 101 (1969). It may, however, prove possible to claim that computer programs are not "original works of authorship" (or even that they are not "writings" in the constitutional sense), for they are poles apart from the literary, artistic or informational works to which copyright traditionally has applied. While the copyright law has gathered a wide variety of items within its scope, ranging from architects' drawings and photographs to code words, computer programs may still prove significantly different in that they are neither of literary or artistic intent nor are they intended to convey information to another person. Cf. Statement of EDUCOM on Copyright Law Revision, 1967 *Senate Hearings*, pt. 2, at 565.

puters know that a program is basically a set of instructions that set a computer's "switches" so that it can perform a particular function; ²³⁴ others may be helped by a simple example. A programmer who wants a computer to calculate how much social security tax to deduct from each worker's pay check will perform two operations. First, he must decide what basic steps the machine will have to go through to make the calculations. He may decide, for example, that the machine will read from each individual's pay card his number, his cumulative social security taxes paid for the year, and his weekly pay; that the machine will then determine if further social security tax is owed and, if so, how much; and that the machine will finally print out the worker's number, the amount of his tax, and his remaining weekly pay. This list of "basic steps" — written out in somewhat greater detail — may be considered the program's "algorithm" or "fundamental idea," which can be embodied in a "flow chart." ²³⁵

²³⁴ Cf. THE PRESIDENT'S COMMISSION ON THE PATENT SYSTEM, "TO PROMOTE THE PROGRESS OF . . . USEFUL ARTS": IN AN AGE OF EXPLODING TECHNOLOGY 12 (1966).

²³⁵ The following is an example of a flow chart for this social security problem. It represents what is probably only one of a number of ways in which a computer program could be written to perform the given task.

SOCIAL SECURITY PROGRAM

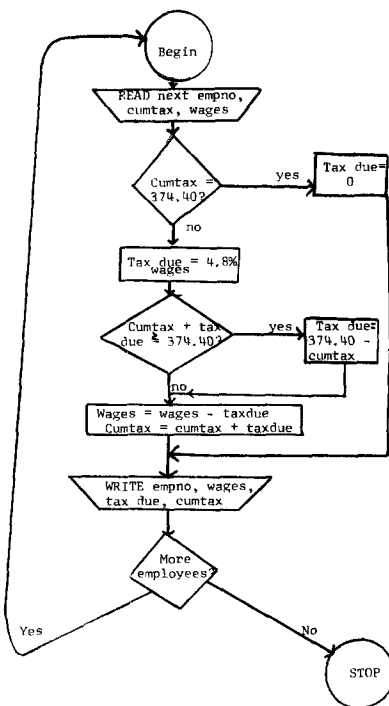
Employee number (empno) =
a four digit number

Cumulative tax (cumtax) =
total tax paid previously
this year.

Wages (input) = salary for
the pay period

Wages (output) = salary
less tax deduction

Tax due = 4.8% of wages
but not to exceed \$374.40



Second, the programmer will translate the algorithm into a set of instructions written in symbolic form that can be inserted into the machine. The machine automatically uses these instructions to set its switches to carry out the desired task. The instructions constitute the "program."²³⁶ Different programs may embody the same algorithm, just as different words can embody the same idea.²³⁷

Programs (also called "software") can be divided very roughly into three sorts. First, *systems programs* control the basic operations of the computing machinery (also called "hardware"). Such a program would, for example, govern the translation of our social security program from a language that vaguely resembles

²³⁶ The following is an example of a program that can be directly inserted into the machine. It is written in a computer-readable language called FORTRAN and represents only one of a number of possible ways of writing a program that embodies the algorithm represented in the flow chart in note 235 *supra*:

```

$JOB                                BREYER/S
1      WRITE (6,22)
2      22 FORMAT (5X,36HSOCIAL SECURITY AND WAGE CALCUL
        1ATION)
3      READ (5,1)LAST
4      1 FORMAT(16)
5      DO 18 J=1, LAST
6      READ (5,2)EMPNO,CUMT,PAY
7      2 FORMAT(I10,2I0.2)
8      UGH=0
9      IF (CUMT-374.40)3,6,6
10     3 UGH=PAY*.048
11     IF ((CUMT+UGH)-374.40)5,4,4
12     4 UGH=374.40-CUMT
13     5 PAY=PAY-UGH
14     CUMT=CUMT+UGH
15     6 WRITE(6,7)EMPNO,PAY,UGH,CUMT
16     7 FORMAT (5X,15HEMPLOYEE NUMBER, 3X,I10/42HWAGES
        1(AFTER SOCIAL SECURITY DEDUCTION = ,2X,F10.2/
        227HSOCIAL SECURITY TAX OWED = ,2X,F10.2/
        338HCUMULATIVE SOCIAL SECURITY PAYMENTS =
        4,2X,F10.2)
17     18 CONTINUE
18     8 WRITE(6,9)LAST
19     9 FORMAT(37HNUMBER OF EMPLOYEES IN CALCULATION
        1-,16)
20     STOP
21     END

```

After inserting the program into the machine and then inserting cards with appropriate information about each employee, the machine will print out information that looks roughly like the following:

EMPLOYEE NUMBER	2726	
WAGES (AFTER SOCIAL SECURITY DEDUCTION)		4177.63
SOCIAL SECURITY TAX OWED		210.64
CUMULATIVE SOCIAL SECURITY PAYMENTS		235.91

²³⁷ For example, the exact order of some of the 21 statements in the program in note 236 *supra* does not make a difference. Nor does it matter (within limits) whether one uses a symbol like UGH or some other symbol to represent, for example, the amount of social security payment. Numerous other variations are possible.

English into "machine language" that directly controls the computer's switches. Second, *application programs* govern the solution of particular problems. Our social security program is one example; a program that allows a computer to operate a steel mill would be another. Most application programs are tailored to fit the needs of individual users. Third, *general purpose application programs* (or "utility programs") are adaptable to meet the needs of many different computer users. A "debugging" program — one that enables the machine to locate mistakes in other programs — is one such, for it can be used by numerous computer operators despite their differing particular interests.²³⁸

To create complex programs, which (unlike our social security program)²³⁹ may contain thousands of statements, can cost thousands of dollars.²⁴⁰ Much of this expenditure pays for the time-consuming job of translating algorithm into finished product.²⁴¹ Yet to copy a finished program may require only a xerox machine.²⁴² Presumably for these reasons, copyright protection — which in principle should prohibit unauthorized copying of the program but allow free use of the algorithm or "idea"²⁴³ — has been advocated as desirable.²⁴⁴ (And, for the moment, I shall assume that the details of a meaningful form of copyright protection could be worked out.)²⁴⁵

²³⁸ This breakdown is used by the International Data Corporation in its recent survey of 200 software users and is adopted here for purposes of convenience in the discussion. See McGovern, *User Software Economics*, in *COMPUTERS-IN-LAW INSTITUTE, NATIONAL LAW CENTER, GEORGE WASHINGTON UNIVERSITY, 1969 PROCEEDINGS: THE LAW OF SOFTWARE L-8* [hereinafter referred to as McGovern, *Software*].

²³⁹ The Social Security Program contains twenty-one statements. See note 236 *supra*.

²⁴⁰ See, e.g., Pantages, *The Problems of Packaged Programs*, 14 *DATAMATION* 75 (April 1968); Reynolds, *Software Protection and Software Sale*, *DATA PROCESSING MAGAZINE* 50 (May 1967); *Software Gap — A Growing Crisis for Computers*, *BUS. WEEK* 126, 132 (Nov. 5, 1966).

²⁴¹ See *Software Writing Called 5% Inspiration and 95% Perspiration by IBM Attorney*, *COMPUTERWORLD* 7 (Aug. 6, 1969).

²⁴² Programs that may cost hundreds of thousands of dollars to create may be copied for as little as fifty dollars. Reynolds, *supra* note 240, at 50.

²⁴³ Copyright thus arguably will not allow its owner to block the free use of the fundamental programming ideas that his programs contain.

²⁴⁴ See, e.g., McOustra, *Legal Protection for Computer Programs*, 8 *THE COMPUTER J.* 289 (1965); McTiernan, *Copyright Protection for Computer Programs*, *J. DATA MANAGEMENT* (Jan. 1966); *UCLA Project* 981-84. See generally 1969 *COMPUTER CONFERENCE, supra* note 238.

²⁴⁵ Whether meaningful copyright protection can be provided is far from clear. The major difficulty arises from the fact that the statements of programs can be so easily altered without changing their function or their efficacy. Unless the courts impose a fairly broad test of what constitutes a "copy," copyright protection may prove of little value to the copyright owner. See Note, *Computer Programs and*

This article, however, has made the general claim that a large difference between the cost of producing a work initially and the cost of copying is not alone sufficient to show that copyright protection is desirable. And the case of computer programs is no exception. One should become suspicious of the need for protection at present upon learning that the software industry is currently burgeoning without the use of copyright²⁴⁶ and that a 1964 announcement by the Copyright Office that it would accept programs for registration has led to the submission of only 200.²⁴⁷ One's suspicions should be further confirmed by the following considerations.

First, much (but not all) *systems* software is now, and should continue to be, created by hardware manufacturers and sold along with their hardware at a single price.²⁴⁸ Many of these programs will work properly only with a particular brand of machine. Both

Proposed Revisions of the Patent and Copyright Laws, 81 HARV. L. REV. 1541, 1549-52 (1968). I suspect, however, that it would be possible to work out a form of meaningful copyright protection.

²⁴⁶ EDP INDUSTRY REPORT, an electronic data processing industry newsletter, estimates that revenues from the sale of contract and packaged software amounted to \$450 million in 1969 and will amount to \$2.5 billion by 1974. 5 EDP INDUSTRY REPORT 3 (March 12, 1970). International Data Corporation estimates that sales by independent software producers have grown from \$12 million to \$320 million in the past six years. See McGovern, *Free Competition and Illegal Restraints in Software*, 1969 COMPUTER CONFERENCE, *supra* note 238, at J-1, J-20 [hereinafter referred to as McGovern, *Competition*]. For the next few years users should continue to produce most programs themselves, and hardware companies will continue to account for a major portion of the remaining software market. IBM's recent promise to price its hardware and its software separately, however, should allow independent software companies to increase considerably their share of this rapidly growing market.

²⁴⁷ Telephone conversation with George Cary, Office of Register of Copyrights, Washington, D.C., Nov. 16, 1970. Of course this number may reflect several other factors. For one thing, programs can be copyrighted without being registered (registration must take place only before an infringement action can be brought). See *Washingtonian Pub. Co. v. Pearson*, 306 U.S. 30 (1939). For another, the law on copyrighting programs has not yet been clarified through court proceedings. Further, programmers may fear that copyright protection will prove ineffectual for the reason stated in note 245 *supra*. Finally, the number of widely distributed programs needing protection may not yet be great.

²⁴⁸ Measured in terms of computer time used, software produced in-house accounted for 54.8% of all software in 1969, software produced by hardware companies accounted for 34.4%, that produced by independent software companies for 5.5%, and that produced by "user libraries" for 5.3%. McGovern, *Competition* J-19. As the result of "unbundling" (IBM's promise to sell hardware and software separately), the share of software produced by hardware companies is expected to fall and that of software independents to rise. But IBM's promise does not apply to *systems* software; such software will still be sold along with IBM hardware at a single price. Doud, *The Business of Software and Its Protection*, 1969 COMPUTER CONFERENCE, *supra* note 238, at P-1, P-8; Goetz, *Future Direction in the Software Industry*, *id.* at M-1 to M-5.

these factors diminish the probability that a copier can sell copies of such software profitably. And a recent survey conducted by the Independent Data Corporation (IDC) indicates that development of systems software in the next few years should account for twenty-five percent of all software development costs.²⁴⁹

Second, most *application* programs (which, according to the IDC, account for about sixty percent of all program development costs)²⁵⁰ are tailored to suit individual customer needs.²⁵¹ In fact, more than half of all computer time is accounted for by programs that computer users develop within their firms for their own use.²⁵² It is unlikely that copying such programs will significantly interfere with their development.

Third, those application programs (including *general purpose application* programs) that are usable generally now ordinarily are sold, not directly "off the shelf," but in "packages." They may come accompanied with copyrighted documentation manuals and a promise that the seller will install the program, iron out its "bugs," update it as advances are made, and make adjustments from time to time to keep it compatible with others in the machine.²⁵³ The cost of these supporting items may equal or exceed the program's initial cost.²⁵⁴ These facts suggest that a computer user is often buying services and expertise as much as he is buying a particular computer program. And as long as this is so, the competitive advantage that a copying seller obtains through the legal right to copy the program itself may be small. In any event, the copier's need to develop this support independently should often provide the initial programmer with sufficient lead time to recover his development costs.

Within the next decade, however, pressures for greater program uniformity and compatibility as well as technological advances may greatly increase the number of generally usable programs sold "off the shelf."²⁵⁵ And independent software companies that may produce such programs are just beginning what may be a period of rapid growth.²⁵⁶ Yet whether copyright pro-

²⁴⁹ McGovern, *Software*, at L-1, L-8.

²⁵⁰ *Id.*

²⁵¹ See McGovern, *Competition*.

²⁵² *Id.* at J-19.

²⁵³ McGovern, *Competition*; McGovern, *Software*; Head & Linick, *Software Package Acquisition*, 14 DATAMATION 22 (Oct. 1968).

²⁵⁴ See, e.g., Goetz, *supra* note 248, at M-2.

²⁵⁵ See *ADR to sell 1000th AUTOFLOW*, 3 COMPUTERWORLD 41 (Dec. 31, 1969); *Software Marketplace Reflects Realignment of Users*, 3 COMPUTERWORLD 83 (Nov. 12, 1969); *Tailor-Made Packages Are Often Unneeded*, 3 COMPUTERWORLD 5 (July 9, 1969).

²⁵⁶ McGovern, *Competition*; McGovern, *Software*.

tection is needed to induce and to support these developments is far from clear. For one thing, the creators of some such programs should be able to recover their development costs by selling copies at fairly high prices to a small number of buyers. In fact, generally usable programs are now often sold to individual users at prices approximating ten to twenty percent of their development costs.²⁵⁷ It should therefore not prove difficult to organize a fairly small group of buyers to pay for the creation of a program they desire.²⁵⁸

More importantly, the next decade or two should see the rapid expansion of "time-sharing" systems, which allow many (perhaps thousands) of subscribers to use a single central group of computers to which they are connected by communications wires.²⁵⁹ For our purposes, it is sufficient to note that such systems can organize automatically large numbers of program users for negotiating purposes (and the members of these systems may do much programming themselves). Copyright protection is not likely to be needed if it is relatively easy to identify, organize, and bargain with groups of potential program buyers. Further, hardware manufacturers — whose programs now account for the use of more than one-third of all computer time²⁶⁰ — may continue to develop some generally usable application programs even if copying threatens recovery of their development costs. Those costs are likely to account for only a very small percentage of their total revenue,²⁶¹ and they may view some of this expense as promotional, since the development of generally usable programs may increase demand for computers. Finally, educational institutions or the government may have a hand in developing certain generally usable programs — particularly those such as "debugging" programs which serve other than purely commercial purposes. The development of such programs is not likely to be affected significantly by the absence of copyright protection.

These considerations cast substantial doubt on the present need for computer program copyright protection and indicate that

²⁵⁷ Pantages, *supra* note 240, at 75.

²⁵⁸ For example, IBM has brought together a group of twelve airlines to pay for its Programmed Airlines Reservations System, which all have adopted with some variations. See Minini, *Implementing the Very Large Applications - Software Package*, 15 DATAMATION 141 (Dec. 1969).

²⁵⁹ Some writers predict the eventual development of a nationwide interconnected time-sharing system. 1970 *Time-Sharing Supplement*, 4 COMPUTERWORLD (Mar. 25, 1970); Fano & Corbato, *Time-Sharing on Computers*, 215 SCIENTIFIC AM. 128 (Sept. 1966).

²⁶⁰ McGovern, *Competition* J-19.

²⁶¹ McGovern estimates that the direct development cost of applications software now runs to 3% or less of the revenue received by computer hardware companies. McGovern, *Competition* J-10.

the case for future need is quite speculative. They show that any such need will focus upon (1) generally usable programs, (2) produced by independent software companies, (3) selling "off the shelf" (4) at low prices (5) to large numbers of widely dispersed buyers. They also suggest that further empirical study might help to measure the real depth of that need by examining possible alternative methods of paying for such program development. Such a study might, for example, help to determine whether the market for "generally usable off the shelf" programs will be a large one, and whether it will consist of a relatively small number of time-sharing networks or whether it is more likely to resemble the market for tradebooks.

In the face of this uncertain need it would seem unwise to extend copyright protection to virtually all computer programs — as the Revision Bill would do — for such an extension may cause considerable harm. For one thing, if computer producers form the habit of copyrighting nearly all their creations (as they may well do once such protection is clearly established), a serious "transaction cost" problem will arise. Many computer users, particularly those belonging to "user groups" or subscribing to "time-sharing" systems, frequently copy portions of the programs of others. In doing so, they may make modifications to improve them or to make them suitable for related uses. As in the case of xeroxing, to obtain permission to copy parts of programs from owners outside the user's own group or time-sharing system will often prove prohibitively expensive and time-consuming. To enforce a "permissions" requirement, in fact, may lead many of these users to borrow only the algorithm of the program part and recreate the program itself, or — what is more likely — to make various changes in the program to avoid the appearance of copying. The former of these alternatives will waste valuable computer and operator time, and the latter, to quote Professor Oettinger, may have "disastrous consequences . . . on standardization in electronic data processing."²⁶²

Moreover, as in the case of books, the freedom from direct competition that copyright provides may allow some program creators to charge prices considerably higher than needed to recover development expenses. The fact that creators of many generally usable programs can now recover their costs after only five or ten sales suggests that this problem may become significant.²⁶³ Further, difficulties in determining whether one program is a "copy" of another may make protection difficult for courts to administer. On the one hand, a court that holds trivial variations

²⁶² 1967 *Senate Hearings* 589.

²⁶³ Pantages, *supra* note 240, at 75.

sufficient to avoid liability would risk emasculating the law. On the other, a court too ready to find "copying" (one, for example, that might predicate liability on "substantial similarity" plus "access to the copyright owner's work") may lead program creators to avoid using even the algorithm of another's program, to write new programs in unusual ways to lessen the risk of "similarity," or to make "similarity searches" through program libraries to which they have had access. One suspects that skillful allocation of burdens of proof may lessen the temptation to engage in these wasteful activities,²⁶⁴ but one cannot be certain, for neither Congress nor the courts have addressed themselves to the problem. Finally, there are anticompetitive possibilities lurking in the fact that a vast number of copyrights would be owned by IBM, which creates a large percentage of all computer programs.²⁶⁵ One must wonder, for example, whether, without protection, smaller hardware or software firms would not find it easier to use parts of IBM programs in their efforts to compete with IBM.²⁶⁶

This sketch of possible harms is not meant to be definitive, but simply to warn against any headlong rush to provide computer programs with protection — particularly fifty-six or seventy-five years of copyright protection — that would apply to virtually all types of computer programs regardless of need.²⁶⁷ This is not to suggest, however, that extending protection through patent²⁶⁸

²⁶⁴ One might impose a heavy burden of proof upon the copyright owner to show that substantial copying has taken place. The owner, might, for example, have to bury various identifying devices within a long program (such as unusual instructions that cancel each other out) to show that a program that looks somewhat different is, in fact, a copy. (Such a possibility might restrict the number of programs receiving meaningful protection, but it would provide protection for some, while minimizing the possibility that copyright will tie up the algorithm.) The defendant might be assumed not to have copied if he can produce the worksheets from which he made his program. Of course, worksheets can be simulated and hidden devices uncovered, but the work of doing so may nearly approximate that of developing a program anew from the basic algorithm. See S. Perlman, *Computer Programs and Proprietary Rights: Defining the Framework*, April 19, 1970 (unpublished thesis in Harvard Law School Library). (Many of the sources cited in this section have been uncovered through Mr. Perlman's research.)

²⁶⁵ See *Id.* at 9, 10; see also notes 246, 248 *supra*.

²⁶⁶ The effect of software copyright on competition in hardware is difficult to predict, but one ought to beware of a form of protection of which IBM appears to be the major beneficiary.

²⁶⁷ Rapidly developing computer technology would make the active life of the ordinary program relatively short. See, e.g., Goetz, *supra* note 248, at M-2.

²⁶⁸ Patent protection applies to novel ideas that are useful and not "obvious . . . to a person having ordinary skill in the art." 35 U.S.C. § 103 (1964). And it prohibits unauthorized use of the patented idea whether or not it has been copied. Patent protection as applied to computer programs would seem of less value, but considerably more danger, than copyright for several reasons. First, the standard of "novelty" is far more difficult to attain than copyright's standard of "origin-

or trade secret²⁶⁹ law is any more desirable. Rather, before ex-

ality;" thus few programs may qualify for protection. Second, since patents tie up ideas, all programmers having new ideas would have to check through existing patents and patent applications to ascertain whether someone else had thought of it first. The expense involved here (and in searching "prior art" to determine whether any conflicting patents are valid) may be so great as to deter program development. Third, the need to "invent around" patented ideas may waste development resources and interfere with standardization of programs. Fourth, patent protection will provide IBM with the power to prevent competitors from using any of the ideas contained in their programs—a far more serious threat to competition than copyright's inhibition on copying them. Finally, patent protection, by imposing a charge upon using an idea, inhibits the flow of already-created ideas throughout the industry. See MACHLUP, *supra* note 149; Note, *Computer Programs and Proposed Revisions of the Patent and Copyright Laws*, 81 HARV. L. REV. 1541, 1552-54 (1968). The Court of Customs and Patent Appeals, however, has indicated a belief that computer programs are patentable. Application of Charles D. Prater, 415 F.2d 1393 (C.C.P.A. 1969); Application of Walter D. Bernhart 417 F.2d 1395 (C.C.P.A. 1969). The view of the patent court has not yet been confirmed by other federal courts. The Patent Reform Act, as submitted to Congress in 1966, provided that computer programs were not patentable, but the section so stating has more recently been removed from the Reform Bill. See Statement of Edward J. Brenner, Commissioner of Patents, in *Hearings on S. 2, S. 1042, S. 1377, S. 1691, S. 2164, S. 2597 before the Subcomm. on Patents, Trademarks, and Copyrights of the Senate Comm. on the Judiciary*, 90th Cong., 1st Sess. 393 (1968).

²⁶⁹ State trade secret law normally protects secret information that gives a firm a competitive advantage from "wrongful" disclosure, by, for example, an industrial spy, an employee under a fiduciary duty not to disclose, or a customer who signs a contract agreeing not to do so. See, e.g., *Developments in the Law—Competitive Torts*, 77 HARV. L. REV. 888, 947-59 (1964). It has been used by many software creators to protect their programs. See, e.g., *Software Gets a Hardsell Approach*, BUS. WEEK 171 (Oct. 21, 1967); *IBM License Agreement*, 1969 COMPUTER CONFERENCE, *supra* note 238. Since trade secret law protects only against wrongful "disclosure," not unauthorized "copying," it creates neither the "transactions cost" problems of copyright nor the inhibitions (and related administrative costs) that the patent law creates for independent rediscovery and use. And since it protects only that information which is, and which remains, secret, its scope is limited, for ideas tend to become public after a time either through leakage or because their creator sees economic gain in their widespread dissemination. In fact, secrecy does not seem to have posed a serious problem for computer programming so far. See Note, *Computer Programs and Proposed Revisions of the Patent and Copyright Laws*, 81 HARV. L. REV. 1541, 1544, 1546-47, 1555 (1968).

Any discussion of the principles underlying trade secret law would be complex, for it is designed not simply to encourage information production, but also to further society's interests in, for example, deterring industrial espionage, protecting the confidential and fiduciary nature of employer/employee relations, and discouraging businesses from devoting resources to elaborate security procedures. And the complexity would be magnified by three recent Supreme Court decisions suggesting that state law protecting trade secrets may be preempted to a considerable extent by federal patent and copyright law. See *Lear v. Atkins*, 395 U.S. 653 (1969); *Sears, Roebuck & Co. v. Stiffel Co.*, 376 U.S. 225 (1964); *Compco Corp. v. Day-Brite Lighting, Inc.*, 376 U.S. 234 (1964). Despite this complexity two points should be mentioned here. The argument is sometimes made that if

tending protection to an industry the size of which may soon dwarf book publishing, a far more careful and extensive canvass of that industry to determine probable need should be made, with particular emphasis upon alternative methods of financing the development of computer programs. And, if need is shown, efforts should be made to tailor protection to minimize the harms that copyright may cause.²⁷⁰ In any event, on the basis of the existing literature on the subject²⁷¹ and the very brief discussion of the matter before Congress,²⁷² it seems fair to conclude that the advocates of copyright protection have not yet made their case.

IV. FINAL CONCLUSIONS

Part II of this article has argued that the general case for copyright protection is weak. Its arguments may prove useful in reaching various specific conclusions about proposed incremental changes in the copyright law. In any event, those arguments lead to at least four policy recommendations:

1. The period of copyright protection is at present too long and should not be extended beyond fifty-six years.
2. The law should allow individuals to make single copies of magazine articles and extracts from books without obtaining permission to do so from copyright owners. The details of this exemption can be determined by the study commission which the Revision Bill would establish.

trade secret protection is granted, patent or copyright protection should also be granted in order to induce disclosure. This argument seems highly dubious as applied to computer programs, however, for difficulties in detecting copyright or patent violations and in enforcing those laws is likely to induce those persons with valuable programs to restrict their dissemination and to seek trade secret protection whether or not they have a patent or copyright option. This factor may explain in part the small number of programs submitted to the copyright office for registration. Second, the case for trade secret protection would seem weakest as applied to programs sold in large numbers to independent customers (under "nondisclosure" contracts). In such cases, the "production inducement" rationale would seem the only principle of protection to apply with force, but, as has been pointed out, the need for protection to induce this sort of production is speculative at best, and protection can prove harmful by inducing program creators to restrict the dissemination of their ideas.

²⁷⁰ One might explore further, for example, the suggestion of EDUCOM that an exclusive right to sell, rather than a right to copy, will suffice. See 1967 *Senate Hearings* 570-73.

²⁷¹ See, e.g., note 244 *supra*. See also 1969 COMPUTER CONFERENCE, *supra* note 238 for further papers on the subject and references to other relevant writings.

²⁷² The references in congressional testimony to computer programs prior to the 1967 Senate hearings (on the basis of a cursory check) seem to be nonexistent. And the 1967 Senate hearings contain only a few scattered references to, and some brief testimony about, the problem. See, e.g., 1967 *Senate Hearings* 198-99, 570-73, 589.

3. The law should allow individuals and small groups to store copyrighted material in computers and to use it for research purposes without obtaining permission from copyright owners.
4. Computer programs should not receive copyright protection at the present time.

The discussion in Part II also has several general implications for the formulation of copyright policy. First, it shows that the argument most frequently advanced to support copyright protection is shallow: To demonstrate that an initial publisher's costs are high, while reproduction costs are low, is not sufficient to establish the need for copyright protection. Rather, one must examine other factors — the probable speed and ferocity of competitive response, the presence of subsidies, the ability of buyers to channel revenue to publishers and authors in the absence of protection — before it can be said that copyright protection is needed.

Second, Part II points out that copyright is very much of a mixed blessing. It can lead to prices higher than necessary to secure production; it can impose large transactions costs; it can even help a firm or group of firms to limit competition throughout an industry.

Third, Part II indicates that the desirability of copyright protection will vary from one type of "writing" to another. One must know facts about the particular industry involved before one can accurately weigh the various costs and benefits associated with copyright protection.

The generally unsatisfactory nature of the congressional hearings on the Copyright Revision Bill underlines the importance of these conclusions. The hearings reveal little critical analysis of industry claims that protection is needed. They show little awareness of the possible harms of extending protection. Rather, the data amassed at the hearings is unsifted, often irrelevant, fact and opinion, and many critical facts about affected industries are missing. Of course the hour is late; the revisors have long been hard at work. Yet one cannot escape the conclusion that more empirical work and more thoughtful analysis is needed before the Copyright Law is significantly revised.

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